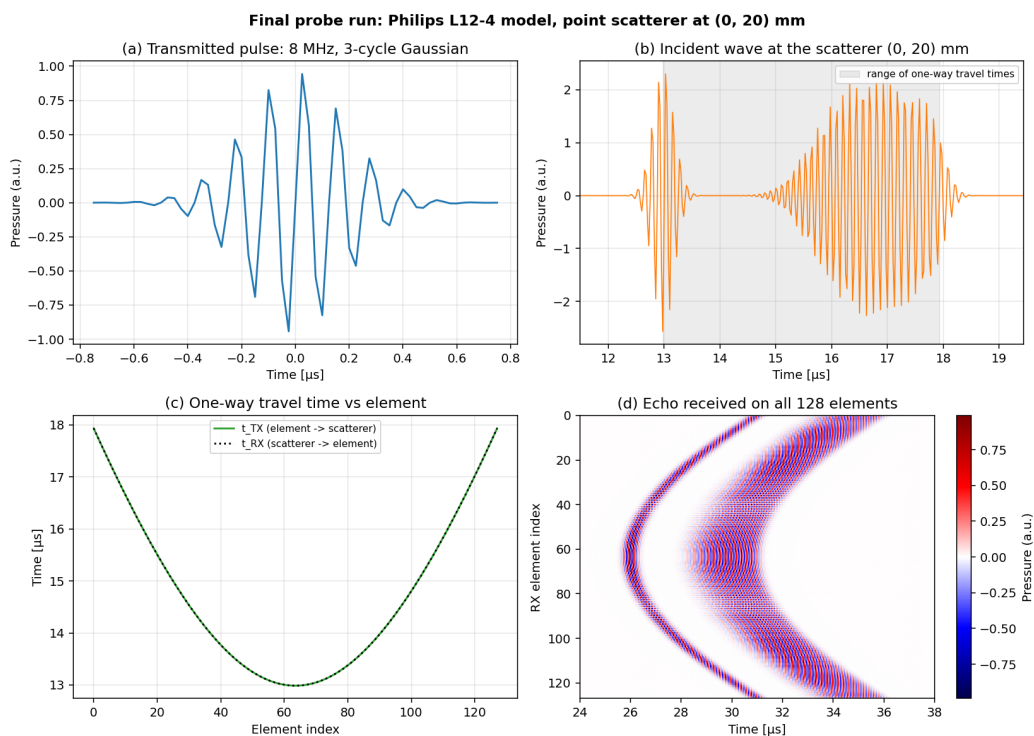


Team 3: Acoustic Propagation

How the sound pulse travels from the probe to a scatterer and back

NITK-UsoundSim · 2-D ultrasound simulator · explanation, input, output and run guide

- The team's code sends a short ultrasound pulse from every element of a linear array, adds up what arrives at a point scatterer, and then sends the echo back to every element.
- It works with a mechanical **pressure wave**: distance, travel time ($t = d / c$) and attenuation (dB/MHz/cm) are the three key ideas.
- One command, `python3 run_team3.py`, runs the 22 unit tests (all pass), the team's own demo (64 elements, 5 MHz) and a second run with the project's final probe (Philips L12-4 model: 128 elements, 0.30 mm pitch, 8 MHz).
- Measured run time on the test Mac: about 2 to 5 seconds.



Final probe run (128 elements, 8 MHz, scatterer at (0, 20) mm). Produced by `run_team3.py`.

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1 Overview

What Team 3 does and what is in the folder

Team 3 builds the **acoustic propagation** part of the NITK-UsoundSim ultrasound simulator. Its job is to answer one question: *when every element of the probe sends a short pulse, what pressure arrives at a point in the tissue, and what pressure comes back to each element?*

The code follows this chain:

- **TX elements** (the array) send a Gaussian pulse.
- **forward_propagation()**: every pulse travels to the scatterer; the pulses are delayed, weakened (attenuated) and added together. This gives the *incident wave* at the scatterer.
- **mock_tissue_interaction()**: a simple stand-in for the tissue: the wave is multiplied by the scatterer's amplitude to give the *reflected wave*.
- **return_propagation()**: the reflected wave travels back to each receive element, delayed and attenuated again. The result is `pressure[element, time]`.

Place in the pipeline

Team 3 sits between transmit beamforming (Team 2) and tissue interaction (Team 4) / receive beamforming (Team 5): it moves the pressure wave from the probe into the tissue and back.

Files in this folder

File / folder	What it is
README.md	Short step-by-step run guide.
requirements.txt	Python packages needed: numpy, matplotlib.
run_team3.py	The one command to run. Added for this demo; it only calls the team's functions.
code/nitk_usoundsim/	The team's ORIGINAL package, copied unchanged (byte-identical).
__init__.py	Marks the folder as a Python package (one docstring line).
config.py	All constants: speed of sound, frequency, sampling rate, array size, phantom points.
acoustic_propagation.py	The main module: pulse, geometry, delays, attenuation, forward and return propagation.
demo_acoustic_propagation.py	The team's demo: makes two figures and two .npz data files.
test_acoustic_propagation.py	22 unit tests.
check_arr.py	Tiny helper that loads a saved .npz file and prints its contents.
requirements.txt	The team's own list of packages.
*.npz, *.png	Outputs the team saved earlier (kept for reference; the new run writes to output/).
input/team3_run_parameters.json	Sample input: array, pulse and scatterer settings for both runs.
output/	Everything made by run_team3.py (plots, data, console log, screenshot).
.vscode/settings.json	VS Code settings for this folder: adds code/ to PYTHONPATH in the VS Code terminal and runs a file from its own folder, so the team's files can be started with the Run button.

2 Theory

The physics behind the code, in simple words

2.1 Sound is a pressure wave

Ultrasound is a mechanical wave: the tissue is squeezed and stretched, so the local pressure goes up and down. The probe element pushes the tissue, the push travels outwards, and later the echo pushes back on the element. Team 3's code works only with this **mechanical pressure** (in arbitrary units, a.u.); turning pressure into an electrical signal is done by other teams.

2.2 Speed of sound, wavelength and pitch

In soft tissue sound travels at about **c = 1540 m/s**. The wavelength is the distance the wave moves in one period of the carrier:

$$\lambda = \frac{c}{f_0} = \frac{1540}{5 \times 10^6} = 0.308 \text{ mm}$$

Team design: $f_0 = 5 \text{ MHz}$. The element pitch is $\lambda/2 = 0.154 \text{ mm}$ (config.py).

A linear array has N elements placed in a straight line, one pitch apart, centred on $x = 0$:

$$x_n = \left(n - \frac{N-1}{2}\right) \cdot \text{pitch}, \quad n = 0, 1, \dots, N-1$$

2.3 Time of flight: $t = d / c$

The element at $(x_n, 0)$ and a scatterer at (x_s, z_s) form a right triangle, so the straight-line distance comes from Pythagoras. Dividing by c gives the travel time:

$$d_n = \sqrt{(x_s - x_n)^2 + z_s^2}, \quad t_n = \frac{d_n}{c}$$

Example (checked by the run)

For the scatterer at (0, 20) mm the centre element is 20 mm away, so $t = 0.020 / 1540 = 12.99 \text{ us}$. With the 128-element L12-4 probe the edge element ($x = 19.05 \text{ mm}$) is 27.62 mm away, so $t = 17.94 \text{ us}$. The echo needs the same time again to come back: round trip = $2z/c = 25.97 \text{ us}$ on axis.

2.4 Attenuation in dB/MHz/cm

Tissue absorbs and scatters energy, so the wave gets weaker with distance, and faster at high frequency. The model uses the usual soft-tissue rule with $\alpha_0 = 0.5 \text{ dB/MHz/cm}$:

$$A_{dB} = \alpha_0 \cdot f_{0, \text{MHz}} \cdot d_{\text{cm}}, \quad A_{\text{lin}} = 10^{-A_{dB}/20}$$

Why divide by **20** and not 10? The decibel is defined on *power*: $\text{dB} = 10 \log_{10}(\text{P ratio})$. Power is proportional to pressure squared, so for a pressure *amplitude* ratio $\text{dB} = 10 \log_{10}(p^2 \text{ ratio}) = 20 \log_{10}(p \text{ ratio})$. The code multiplies pressure amplitudes, so it must use /20.

Example

5 MHz, 2 cm: $A = 0.5 \times 5 \times 2 = 5 \text{ dB}$, so the amplitude factor is $10^{(-5/20)} = 0.562$ one way. There and back: 10 dB, factor 0.316 (these are exactly the numbers the run prints for $z = 20 \text{ mm}$).

2.5 The transmitted pulse

Each element sends a short burst: a sine wave at the centre frequency f_0 , shaped by a bell-shaped (Gaussian) envelope so that it starts and ends smoothly:

$$p(t) = \sin(2\pi f_0 t) \exp\left(-\frac{t^2}{2\sigma^2}\right), \quad \sigma = \frac{N_{\text{cycles}}}{2f_0}$$

With $N_{\text{cycles}} = 3$: at 5 MHz $\sigma = 0.3$ us; at 8 MHz $\sigma = 0.1875$ us. `generate_pulse()` samples the pulse from -4σ to $+4\sigma$ (the envelope is about e^{-8} there, practically zero). A shorter pulse gives better axial (depth) resolution.

2.6 Forward propagation: a sum over elements

All elements fire together (no focusing: the extra transmit delay τ_{TX} is zero). The pressure at the scatterer is the **sum** of all the delayed, attenuated pulses:

$$p_{\text{inc}}(t) = \sum_{n=0}^{N-1} A_{TX}(n) p(t - \tau_{TX}(n) - t_{TX}(n))$$

Because $p(t)$ is a formula, the code evaluates it exactly at the shifted time; no interpolation is needed.

2.7 Return propagation: shifting a sampled signal

The reflected wave is now a list of numbers (samples), not a formula. To delay it by $t_{RX}(m)$ for receive element m , the code reads the reflected wave at the times $t - t_{RX}(m)$ using **linear interpolation** (`np.interp`): between two samples it draws a straight line. Times before the start or after the end are set to 0.

$$p_{RX,m}(t) = A_{RX}(m) p_{\text{refl}}(t - t_{RX}(m))$$

2.8 Linear superposition

The model is linear: waves simply add. With several scatterers, each echo is computed on its own and the received signals are added. A unit test checks that the total equals the sum of the parts.

2.9 Sampling and Nyquist

Signals are stored at $f_s = 40$ MHz, one sample every 25 ns. The Nyquist rule says f_s must be more than twice the highest frequency present. For a 5 MHz or 8 MHz pulse (bandwidth reaching roughly 12 MHz at 8 MHz) 40 MHz is enough, giving 8 (5 MHz) or 5 (8 MHz) samples per carrier period.

2.10 Simplifications of this model

- Sound goes in **straight lines** at a **constant speed** (no refraction, one tissue type).
- Each element is a **point**: no element width, no directivity, no apodization.
- **No 1/r spreading**: only attenuation reduces the amplitude, not the geometric spreading of the wave.
- The tissue is a **placeholder**: `mock_tissue_interaction()` just multiplies by 'amp'. No impedance, reflection coefficient or speckle.
- Attenuation is applied at the centre frequency only (no change of pulse shape with depth).
- No transmit focusing by default ($\tau_{tx} = 0$), linear physics only.

3 Code explanation

Every file, section by section

3.1 config.py: the single source of truth

config.py holds all constants that other modules must import instead of re-typing them:

Name	Value	Meaning
C	1540.0 m/s	speed of sound in tissue
ALPHA_0	0.5 dB/MHz/cm	attenuation coefficient
F0	5.0e6 Hz	centre frequency
FS	40.0e6 Hz	sampling frequency
N_ELEMENTS	64	number of array elements
LAMBDA, PITCH	$C/F0 = 0.308$ mm, $LAMBDA/2 = 0.154$ mm	wavelength and element spacing
X_MIN..X_MAX, Z_MIN..Z_MAX	-10..10 mm, 0..40 mm	imaging field of view
N_SCANLINES	64	number of image lines (used by later teams)
PHANTOM_POINTS	(0,10), (0,20), (0,30) mm, amp 1.0	three test scatterers
DYNAMIC_RANGE_DB, OUTPUT_SIZE	50 dB, (512, 512)	display settings (used by later teams)

3.2 acoustic_propagation.py: function by function

The top docstring describes the signal flow and units (metres, seconds, Hz). All constants are imported from config.py. $N_CYCLES = 3$ is the one design choice about pulse length.

Function	What it does, in plain words
<code>_pulse_shape(t, f0, n_cycles)</code>	The pulse formula $p(t) = \sin(2 \pi f_0 t) \exp(-t^2 / 2 \sigma^2)$ with $\sigma = n_cycles / (2 f_0)$. It works for any time value, so a delayed pulse is just <code>_pulse_shape(t - delay)</code> . The leading underscore means it is an internal helper.
<code>generate_pulse(fs, f0, n_cycles)</code>	Makes a sampled copy of the pulse for plotting: a time axis from -4σ to $+4 \sigma$ in steps of $1/fs$, and $p(t)$ on it. Returns (t, p). Propagation does not use this array; it uses <code>_pulse_shape</code> directly.
<code>create_linear_array(n_elements, pitch)</code>	Returns the x-position of every element: $(index - (N-1)/2) \times pitch$, so the array is centred on 0. Element 0 is at the most negative x.
<code>calculate_tx_distance(element_x, x_s, z_s)</code>	Pythagoras distance $\sqrt{(x_s - x_n)^2 + z_s^2}$ from each transmit element to the scatterer.
<code>calculate_rx_distance(element_x, x_s, z_s)</code>	The same formula for the way back (scatterer to each receive element). It is kept as a separate function so receive-only effects can be added later without touching the transmit path.
<code>calculate_propagation_delay(distance, c)</code>	Travel time $t = d / c$.
<code>calculate_attenuation(distance, f0, alpha0)</code>	Converts Hz to MHz and m to cm, computes the loss in dB = $\alpha_0 \times f_MHz \times d_cm$ and returns the linear amplitude factor $10^{(-dB/20)}$, a number between 0 and 1.

Function	What it does, in plain words
<code>forward_propagation(scatterer, element_x, t_axis, ...)</code>	Computes distances, times and attenuation for every element, sets <code>tau_tx</code> to zeros if not given (a hook for a future transmit-focusing team), and then loops over the elements adding $A_{TX}(n) \times p(t - \tau_{tx}(n) - t_{TX}(n))$. Returns (<code>incident_wave</code> , <code>d_tx</code> , <code>t_tx</code>).
<code>mock_tissue_interaction(incident_wave, scatterer)</code>	Temporary placeholder for Team 4: returns <code>incident_wave</code> x <code>scatterer['amp']</code> . Same length in and out, so the real tissue module can be plugged in later without changing <code>return_propagation()</code> .
<code>return_propagation(reflected_wave, t_axis, scatterer, element_x, ...)</code>	For every receive element <code>m</code> it computes <code>d_RX</code> , <code>t_RX</code> and <code>A_RX</code> , then shifts the sampled reflected wave with <code>np.interp(t_axis - t_RX(m), t_axis, reflected_wave, left=0, right=0)</code> and multiplies by <code>A_RX(m)</code> . Returns (<code>pressure[element, time]</code> , <code>d_rx</code> , <code>t_rx</code>).
<code>simulate_acoustic_propagation(phantom_points, ...)</code>	The whole pipeline. (1) builds the array and pulse; (2) finds the longest round trip over all scatterers and builds a shared time axis from minus the pulse half-width to that time plus a margin (15 % of the round trip + half-width); (3) for each scatterer runs forward -> mock tissue -> return and adds the result to the total (linear superposition); (4) returns a dictionary with <code>t_axis</code> , <code>pressure</code> (and its alias <code>received_pressure</code>), <code>incident_pressure</code> , <code>reflected_pressure</code> , <code>element_x</code> , the pulse and a <code>per_scatterer</code> list with all the delays and partial signals.

3.3 demo_acoustic_propagation.py

- Imports `config` and `acoustic_propagation` (with a fall-back import if run from inside the package).
- OUT_DIR** is the folder of the demo file itself: by default the demo saves next to its own source. `run_team3.py` instead passes `out_dir=output/` so the original folder is not touched.
- make_figures(out_dir)** runs `simulate_acoustic_propagation()` and draws Figure 1 (array + scatterers, pulse, delay vs depth, attenuation vs depth) and Figure 2 (incident waves, reflected waves, the waveform on RX element #32, and the element-by-time image).
- save_part_outputs(result, out_dir)** saves two self-describing .npz files: PART 1 (forward propagation: incident pressure at each scatterer) and PART 2 (return propagation: received pressure on all 64 elements). Every array has labels and units stored next to it.
- The `__main__` block calls both functions and prints the file names and array shapes.

3.4 test_acoustic_propagation.py (22 tests)

Test class	What it checks
<code>TestArrayGeometry</code> (4)	element count, spacing = pitch, array centred on 0, $LAMBDA = C/F0$ and $PITCH = LAMBDA/2$
<code>TestDistancesAndDelays</code> (7)	TX and RX distance formulas, on-axis distance = depth, $t = d/c$ for TX and RX, deeper = later, off-axis target gives different delays
<code>TestAttenuation</code> (4)	matches the dB formula, 20 mm at 5 MHz = 5 dB, deeper = weaker, factor in (0, 1]
<code>TestPulse</code> (2)	pulse sampled at $1/FS$, peak near $t = 0$
<code>TestFullPipeline</code> (5)	output shape, real and finite, 20 mm round trip = 25.97 us (50 ns tolerance), deeper points later, superposition

3.5 check_arr.py and __init__.py

`check_arr.py` is a small scratch script: it loads

`nitk_usoundsim/part2_return_propagation_outputs.npz` (path relative to the code/ folder) and prints the description, the reflected pressure array, its shape and row labels. Two other loads are commented out. `run_team3.py` does not call it. `__init__.py` contains only a docstring.

3.6 How run_team3.py uses the team's own code

In short

All the acoustic propagation logic (array, pulse, distances, delays, attenuation, forward and return propagation, the full pipeline, the demo figures and the 22 tests) comes from the team's existing code in `code/nitk_usoundsim/`. That code is used **unchanged**; it was not rewritten.

`run_team3.py` is a **demo wrapper**: it imports the team's modules, calls their functions with chosen arguments, and prints and saves the results.

Where run_team3.py uses your code

run_team3.py line(s)	Team file / function used	What it does there
44, 45, 46	<code>config</code> , <code>acoustic_propagation</code> as <code>ap</code> , <code>demo_acoustic_propagation</code> as <code>demo</code>	Imports the team's three modules (line 37 first adds <code>code/</code> to the Python path).
109-111	<code>test_acoustic_propagation.py</code> (all 22 unit tests)	Runs <code>python -m unittest nitk_usoundsim.test_acoustic_propagation -v</code> in a separate process inside <code>code/</code> (Step 1).
141-144	<code>config.N_ELEMENTS</code> , <code>F0</code> , <code>FS</code> , <code>C</code> , <code>ALPHA_0</code> , <code>PHANTOM_POINTS</code>	Compares the team's constants with the values listed in the input JSON (Step 2).
146-154	<code>config.N_ELEMENTS</code> , <code>F0</code> , <code>FS</code> , <code>C</code> , <code>LAMBDA</code> , <code>PITCH</code> , <code>ALPHA_0</code> , <code>PHANTOM_POINTS</code>	Prints the team's design values.
158	<code>demo.make_figures(out_dir=OUT_DIR)</code>	Runs the team's demo: it calls <code>ap.simulate_acoustic_propagation()</code> and draws the team's two figures, saved in <code>output/</code> .
159	<code>demo.save_part_outputs(result, out_dir=OUT_DIR)</code>	Saves the team's PART 1 and PART 2 .npz files in <code>output/</code> .
177	<code>ap.calculate_attenuation(d)</code>	One-way amplitude factor for each demo scatterer.
180	<code>config.C</code>	Speed of sound for the $2z/c$ check value.
194	<code>ap.create_linear_array(n_el, pitch)</code>	Element positions of the 128-element final probe (Step 3).
195	<code>ap.generate_pulse(fs=fs, f0=f0)</code>	The 8 MHz pulse; its last time value is used as the half-width.
200-203	<code>ap.calculate_tx_distance</code> , <code>ap.calculate_rx_distance</code> , <code>ap.calculate_propagation_delay</code>	Travel times, used here only to find how long the time axis must be.
208	<code>ap.forward_propagation(scat, element_x, t_axis, f0=f0)</code>	Incident wave at the scatterer (0, 20) mm.
209	<code>ap.mock_tissue_interaction(incident, scat)</code>	Reflected wave (incident x amp).
210	<code>ap.return_propagation(reflected, t_axis, scat, element_x, f0=f0)</code>	Received pressure on all 128 elements.
215, 216, 243	<code>ap.N_CYCLES</code>	Number of pulse cycles, printed and used in a figure title.
226	<code>ap.calculate_attenuation(d_tx, f0=f0)</code>	One-way amplitude factor at 8 MHz for every element.
296	<code>config.ALPHA_0</code>	Printed at the top of Step 4.
299, 302	<code>ap.calculate_attenuation(depth, f0=f)</code> and <code>(0.020, f0=f)</code>	Attenuation vs depth at 4, 8 and 12 MHz, and the value at 20 mm (Step 4).

What run_team3.py adds on top (written for this demo, not the team's code)

run_team3.py line(s)	Demo addition
31, 33-37, 39-40	Set-up: no <code>__pycache__</code> in code/, folder paths, add code/ to the Python path, matplotlib without windows (Agg).
52-98	Helpers: Tee (copies printed text into a log), banner, rel, and console_to_png (draws the console text as a picture).
104-132	Step 1 wrapper: saves the full unittest text, prints one short line per test and counts the passes.
145, 166-180	Step 2 extra printout. Own calculations: round trip = $\min t_{TX} + \min t_{RX}$ (line 176), two-way amplitude = $\text{calculate_attenuation}(\dots)$ squared (177), and the check value $2z/c$ (180).
196-206	Builds the shared time axis for the final probe run. It copies the rule used inside the team's <code>simulate_acoustic_propagation()</code> : start at minus the pulse half-width, end at max round trip + 15 % of it + half-width, step $1/f_s$. It is needed because Step 3 calls the individual functions, not the full pipeline.
212-237	Step 3 printout. Own calculations: aperture (213), $\sigma = N_{CYCLES} / (2 f_0)$ (216), round-trip times $t_{TX} + t_{RX}$ (224-225), $\text{dB} = -20 \log_{10}(\text{factor})$ (228), and peak search with <code>np.argmax</code> (229-237).
240-285	Extra figure <code>L12-4_final_probe_propagation.png</code> (four panels) and <code>L12-4_final_probe_outputs.npz</code> .
291-316	Step 4: depth grid <code>np.linspace(0, 60 mm, 301)</code> (294). Own formulas: one-way $\text{dB} = -20 \log_{10}(a)$ (300, 303), round-trip factor = a squared (301, 304), round-trip $\text{dB} = -40 \log_{10}(a)$ (304). Saves <code>attenuation_vs_depth_4_8_12MHz.png</code> .
329-334	The step sequence: reads <code>input/team3_run_parameters.json</code> (329-330), then runs Steps 1 to 4 in order (331-334).
335-352	Summary and file list, then <code>output/console_log.txt</code> and <code>output/console_screenshot.png</code> .

To see it yourself: open `run_team3.py` at the lines above; Cmd/Ctrl+Click a team function in VS Code opens the original file in code/.

3.7 run_team3.py (added for this demo)

- Adds code/ to the Python path, reads `input/team3_run_parameters.json` and copies all printed text into a log.
- Step 1: runs the unit tests in a separate process inside code/ and counts the passes.
- Step 2: calls the demo's `make_figures()` and `save_part_outputs()` with `out_dir = output/`.
- Step 3: final probe run by arguments only: `create_linear_array(128, 0.30e-3)`, `generate_pulse(f0=8e6)`, a 40 MHz time axis from minus the pulse half-width, `forward_propagation(..., f0=8e6)`, `mock_tissue_interaction`, `return_propagation(..., f0=8e6)`; prints key numbers, saves a figure and an `.npz`.
- Step 4: `calculate_attenuation` at 4, 8 and 12 MHz vs depth.
- Finally writes `console_log.txt` and a screenshot-style `console_screenshot.png`.

3.8 Full code listings

code/nitk_usoundsim/__init__.py

```
1 """NITK-UsoundSim - Acoustic Propagation module (Phase 1)."""
```

code/nitk_usoundsim/config.py

```
1 """
```

```

2 config.py
3 =====
4
5 Global configuration for NITK-UsoundSim.
6
7 This is the SINGLE SOURCE OF TRUTH for medium, transducer, imaging-FOV,
8 phantom and display parameters. Every other module (acoustic propagation,
9 tissue interaction, beamforming, image formation, ...) MUST import these
10 values from here rather than redefining or hard-coding them.
11 """
12
13 import numpy as np
14
15 # -----
16 # Medium parameters
17 # -----
18 C = 1540.0          # speed of sound in tissue [m/s]
19 ALPHA_0 = 0.5       # attenuation coefficient [dB / MHz / cm]
20
21 # -----
22 # Transducer array
23 # -----
24 F0 = 5.0e6          # transducer centre frequency [Hz]
25 FS = 40.0e6         # sampling frequency [Hz]
26 N_ELEMENTS = 64     # number of array elements
27 LAMBDA = C / F0      # wavelength in tissue at F0 [m]
28 PITCH = LAMBDA / 2   # element pitch (lambda/2 spacing) [m]
29
30 # -----
31 # Imaging field of view
32 # -----
33 X_MIN, X_MAX = -0.010, 0.010 # lateral FOV [m]
34 Z_MIN, Z_MAX = 0.000, 0.040  # axial (depth) FOV [m]
35 N_SCANLINES = 64
36
37 # -----
38 # Calibration targets (synthetic point phantom)
39 # -----
40 PHANTOM_POINTS = [
41     {"x": 0.000, "z": 0.010, "amp": 1.0},
42     {"x": 0.000, "z": 0.020, "amp": 1.0},
43     {"x": 0.000, "z": 0.030, "amp": 1.0},
44 ]
45
46 DYNAMIC_RANGE_DB = 50.0
47 OUTPUT_SIZE = (512, 512)

```

code/nitk_usoundsim/acoustic_propagation.py

```

1 """
2 acoustic_propagation.py
3 =====
4
5 Acoustic Propagation module for NITK-UsoundSim (Phase 1).
6
7 Implements a SIMPLE, FIELD-II-STYLE LINEAR SPATIAL IMPULSE RESPONSE MODEL
8 for a 64-element linear array. This is an educational simplification of
9 the real spatial-impulse-response formalism used by Field II, not a
10 full acoustic-field simulator (no element directivity/apodization,
11 no diffraction-limited beam profile, no nonlinear/pseudospectral
12 propagation). Those belong to a later, more realistic propagation model.
13
14 Signal flow (this module owns everything from the TX pulse up to the
15 mechanical pressure arriving back at the 64 receive elements)::
16
17     64 TX elements
18     |
19     forward_propagation()      -> incident mechanical wave at scatterer
20     |
21     mock_tissue_interaction()  -> reflected mechanical wave (TEMPORARY,
22     |                               swappable stand-in for the real
23     |                               Tissue Interaction module)
24     return_propagation()       -> pressure[64 RX elements, time]
25
26 The data stays a MECHANICAL ACOUSTIC PRESSURE WAVEFORM throughout - this
27 module never touches RF/electrical channel data (that is a later stage,

```

```

28 e.g. transducer receive sensitivity + electronics, owned by another
29 module).
30
31 Units
32 -----
33 - Distances: metres.
34 - Time: seconds.
35 - Frequency: Hz internally; converted to MHz only where ALPHA_0's
36 dB/MHz/cm convention requires it.
37 - ALPHA_0 is a dB/MHz/cm *amplitude* attenuation coefficient (see
38   calculate_attenuation for the exact dB -> linear-amplitude conversion).
39
40 All physical constants come from ``nitk_usoundsim.config`` and are never
41 redefined here.
42 """
43
44 from typing import Dict, List, Optional, Tuple
45
46 import numpy as np
47
48 from nitk_usoundsim.config import (
49     ALPHA_0,
50     C,
51     F0,
52     FS,
53     N_ELEMENTS,
54     PHANTOM_POINTS,
55     PITCH,
56 )
57
58 # -----
59 # Assumption: transmitted pulse shape
60 # -----
61 # The project brief leaves the exact pulse duration/envelope to be chosen
62 # reasonably. We use a Gaussian-modulated sinusoid
63 #
64 #  $p(t) = \sin(2\pi F_0 t) * \exp(-t^2 / (2\sigma^2))$ 
65 #
66 # with sigma set so the envelope spans approximately N_CYCLES periods of
67 # the F0 carrier (sigma = N_CYCLES / (2 * F0)). N_CYCLES = 3 is a typical
68 # choice for a diagnostic-imaging-like transmit pulse: short enough for
69 # reasonable axial resolution, long enough to have a well-defined centre
70 # frequency. This is documented here as the single assumption governing
71 # pulse shape; change N_CYCLES to explore other pulse lengths.
72 N_CYCLES = 3
73
74
75 def _pulse_shape(t: np.ndarray, f0: float = F0, n_cycles: int = N_CYCLES) -> np.ndarray:
76     """
77     Closed-form Gaussian-modulated sinusoidal pulse, centred at t = 0.
78
79     Physical meaning: the mechanical pressure waveform emitted by a
80     transducer element, evaluated at arbitrary time(s) ``t``. Because
81     this is a closed-form function of time (not a fixed sample array),
82     it can be evaluated directly at any propagation-delayed time
83     (t - t_TX(n)) without needing to resample/interpolate – this keeps
84     forward propagation numerically exact regardless of delay size.
85
86     Parameters
87     -----
88     t : array_like
89         Time(s) [s] at which to evaluate the pulse, relative to the
90         pulse centre (t=0 is the envelope peak).
91     f0 : float
92         Carrier (centre) frequency [Hz].
93     n_cycles : int
94         Approximate number of carrier cycles under the Gaussian envelope.
95
96     Returns
97     -----
98     np.ndarray
99         Pressure amplitude (arbitrary units, peak ~ 1) at each time in t.
100     """
101     sigma = n_cycles / (2.0 * f0)
102     envelope = np.exp(-(t ** 2) / (2.0 * sigma ** 2))
103     return np.sin(2.0 * np.pi * f0 * t) * envelope
104

```

```

105
106 def generate_pulse(
107     fs: float = FS, f0: float = F0, n_cycles: int = N_CYCLES
108 ) -> Tuple[np.ndarray, np.ndarray]:
109     """
110     Generate a short, sampled Gaussian-modulated sinusoidal TX pulse for
111     display/reference purposes (e.g. plotting "the transmitted pulse").
112
113     Internally, forward/return propagation do NOT resample this array;
114     they re-evaluate ``_pulse_shape`` directly at whatever delayed time
115     is needed. This function exists to give a concrete, sampled
116     ``p(t)`` waveform matching the brief's example::
117
118         p(t) = sin(2*pi*f0*t) * Gaussian_envelope(t)
119
120     Parameters
121     -----
122     fs : float
123         Sampling frequency [Hz] (defaults to the global FS).
124     f0 : float
125         Carrier frequency [Hz] (defaults to the global F0).
126     n_cycles : int
127         Number of carrier cycles under the envelope (see N_CYCLES).
128
129     Returns
130     -----
131     t : np.ndarray
132         Time axis [s], centred at 0.
133     p : np.ndarray
134         Sampled pulse amplitude at each time in ``t``.
135     """
136     sigma = n_cycles / (2.0 * f0)
137     t_half_width = 4.0 * sigma # generous support: envelope is ~e^-8 there
138     t = np.arange(-t_half_width, t_half_width, 1.0 / fs)
139     p = _pulse_shape(t, f0=f0, n_cycles=n_cycles)
140     return t, p
141
142
143 def create_linear_array(
144     n_elements: int = N_ELEMENTS, pitch: float = PITCH
145 ) -> np.ndarray:
146     """
147     Compute the lateral (x) position of every element of a linear array,
148     centred on x = 0 (the array's geometric centre defines the imaging
149     origin, consistent with the symmetric X_MIN/X_MAX FOV in config.py).
150
151     Parameters
152     -----
153     n_elements : int
154         Number of elements (defaults to the global N_ELEMENTS).
155     pitch : float
156         Centre-to-centre element spacing [m] (defaults to the global PITCH).
157
158     Returns
159     -----
160     np.ndarray, shape (n_elements,)
161         x-position [m] of each element, element 0 at the most negative x.
162     """
163     indices = np.arange(n_elements) - (n_elements - 1) / 2.0
164     return indices * pitch
165
166
167 def calculate_tx_distance(
168     element_x: np.ndarray, x_s: float, z_s: float
169 ) -> np.ndarray:
170     """
171     One-way distance from transmit element(s) to a scatterer at (x_s, z_s).
172
173     
$$d_{TX}(n) = \sqrt{(x_s - x_n)^2 + z_s^2}$$

174
175     Parameters
176     -----
177     element_x : array_like
178         TX element x-position(s) [m].
179     x_s, z_s : float
180         Scatterer lateral / axial (depth) position [m].
181

```

```

182     Returns
183     -----
184     np.ndarray
185         Distance [m] from each TX element to the scatterer.
186     """
187     element_x = np.asarray(element_x, dtype=float)
188     return np.sqrt((x_s - element_x) ** 2 + z_s ** 2)
189
190
191 def calculate_rx_distance(
192     element_x: np.ndarray, x_s: float, z_s: float
193 ) -> np.ndarray:
194     """
195     One-way distance from a scatterer at (x_s, z_s) to receive element(s).
196
197      $d_{RX}(m) = \sqrt{(x_s - x_m)^2 + z_s^2}$ 
198
199     Geometrically identical to calculate_tx_distance (the array is
200     reciprocal - same elements act as TX and RX here); kept as a
201     separate, explicitly-named function so TX/RX paths stay independently
202     readable/testable and so RX-specific effects (e.g. receive-aperture
203     truncation, element directivity) can be added later without touching
204     the TX path.
205
206     Parameters
207     -----
208     element_x : array_like
209         RX element x-position(s) [m].
210     x_s, z_s : float
211         Scatterer lateral / axial (depth) position [m].
212
213     Returns
214     -----
215     np.ndarray
216         Distance [m] from the scatterer to each RX element.
217     """
218     element_x = np.asarray(element_x, dtype=float)
219     return np.sqrt((x_s - element_x) ** 2 + z_s ** 2)
220
221
222 def calculate_propagation_delay(distance: np.ndarray, c: float = C) -> np.ndarray:
223     """
224     Convert a one-way propagation distance to a propagation time.
225
226      $t = d / C$ 
227
228     Parameters
229     -----
230     distance : array_like
231         Propagation distance [m].
232     c : float
233         Speed of sound [m/s] (defaults to the global C).
234
235     Returns
236     -----
237     np.ndarray
238         Propagation time [s].
239     """
240     return np.asarray(distance, dtype=float) / c
241
242
243 def calculate_attenuation(
244     distance: np.ndarray, f0: float = F0, alpha0: float = ALPHA_0
245 ) -> np.ndarray:
246     """
247     One-way amplitude attenuation factor for a mechanical pressure wave.
248
249     ALPHA_0 is given in dB / MHz / cm (a standard soft-tissue attenuation
250     convention). To use it we must convert:
251
252     - frequency: Hz -> MHz    (divide by 1e6)
253     - distance:  m  -> cm     (multiply by 100)
254
255     then the attenuation in dB is
256
257      $atten_{dB} = ALPHA_0 * f0_{MHz} * distance_{cm}$ 
258

```

```

259     and the dB -> linear AMPLITUDE conversion (not power, hence /20 and
260     not /10) is
261
262     atten_linear = 10 ** (-atten_dB / 20)
263
264     Parameters
265     -----
266     distance : array_like
267         One-way propagation distance [m].
268     f0 : float
269         Frequency at which attenuation is evaluated [Hz] (defaults to F0).
270     alpha0 : float
271         Attenuation coefficient [dB/MHz/cm] (defaults to the global ALPHA_0).
272
273     Returns
274     -----
275     np.ndarray
276         Dimensionless linear amplitude attenuation factor in (0, 1].
277     """
278     distance = np.asarray(distance, dtype=float)
279     f0_mhz = f0 / 1.0e6 # Hz -> MHz
280     distance_cm = distance * 100.0 # m -> cm
281     atten_db = alpha0 * f0_mhz * distance_cm
282     atten_linear = 10.0 ** (-atten_db / 20.0)
283     return atten_linear
284
285
286 def forward_propagation(
287     scatterer: Dict[str, float],
288     element_x: np.ndarray,
289     t_axis: np.ndarray,
290     f0: float = F0,
291     c: float = C,
292     alpha0: float = ALPHA_0,
293     n_cycles: int = N_CYCLES,
294     tau_tx: Optional[np.ndarray] = None,
295 ) -> Tuple[np.ndarray, np.ndarray, np.ndarray]:
296     """
297     Forward propagation: transducer (all TX elements) -> scatterer.
298
299     The incident pressure field AT THE SCATTERER is the linear
300     superposition of the (delayed, attenuated) pulse contributions
301     arriving from every one of the N_ELEMENTS transmit elements -
302     exactly the "incident mechanical wave" in the module diagram.
303
304     d_TX(n) = sqrt((x_s - x_n)^2 + z_s^2)
305     t_TX(n) = d_TX(n) / C
306     incident_wave(t) = sum_n A_TX(n) * p(t - tau_TX(n) - t_TX(n))
307
308     ``tau_tx`` is an explicit hook for a future transmit
309     beamforming/focusing delay (t_total = tau_TX(n) + t_TX(n) + t_RX(m)).
310     It defaults to all-zeros ("zero/default transmit delays"), as
311     instructed - no focusing implementation is invented here.
312
313     Parameters
314     -----
315     scatterer : dict
316         Must contain "x" and "z" (scatterer position [m]). "amp" is not
317         used here - amplitude is applied by mock_tissue_interaction().
318     element_x : array_like, shape (n_elements,)
319         TX element x-positions [m] (from create_linear_array()).
320     t_axis : array_like, shape (n_samples,)
321         Shared simulation time axis [s] the incident wave is evaluated on.
322     f0, c, alpha0, n_cycles : float / int
323         Physical parameters, default to the global config values.
324     tau_tx : array_like, optional
325         Per-element transmit delay [s] (beamforming/focusing). Defaults
326         to zeros (no focusing) if not supplied.
327
328     Returns
329     -----
330     incident_wave : np.ndarray, shape (n_samples,)
331         Incident mechanical pressure waveform at the scatterer, sampled
332         on t_axis.
333     d_tx : np.ndarray, shape (n_elements,)
334         One-way TX distance for each element (returned for diagnostics/plots).
335     t_tx : np.ndarray, shape (n_elements,)

```

```

336     One-way TX propagation delay for each element (diagnostics/plots).
337     """
338     element_x = np.asarray(element_x, dtype=float)
339     t_axis = np.asarray(t_axis, dtype=float)
340
341     d_tx = calculate_tx_distance(element_x, scatterer["x"], scatterer["z"])
342     t_tx = calculate_propagation_delay(d_tx, c=c)
343     atten_tx = calculate_attenuation(d_tx, f0=f0, alpha0=alpha0)
344
345     if tau_tx is None:
346         tau_tx = np.zeros_like(t_tx)
347     else:
348         tau_tx = np.asarray(tau_tx, dtype=float)
349
350     incident_wave = np.zeros_like(t_axis)
351     for n in range(element_x.size):
352         delay_n = tau_tx[n] + t_tx[n]
353         incident_wave += atten_tx[n] * _pulse_shape(
354             t_axis - delay_n, f0=f0, n_cycles=n_cycles
355         )
356
357     return incident_wave, d_tx, t_tx
358
359
360 def mock_tissue_interaction(
361     incident_wave: np.ndarray, scatterer: Dict[str, float]
362 ) -> np.ndarray:
363     """
364     TEMPORARY synthetic Tissue Interaction interface (Part 2).
365
366     The real Tissue Interaction module (owned by another team) is not
367     ready yet, so this stand-in models the scatterer as simply scaling
368     the incident mechanical wave by its "amp":
369
370         reflected_wave = incident_wave * scatterer["amp"]
371
372     No acoustic impedance, reflection coefficients, or speckle physics
373     are modelled - this is intentionally the simplest possible
374     placeholder. The function's contract (incident wave in, reflected
375     wave out, same length/sampling) is what return_propagation() relies
376     on, so a real Tissue Interaction module can later be swapped in here
377     without changing return_propagation() at all.
378
379     Parameters
380     -----
381     incident_wave : np.ndarray
382         Incident mechanical pressure waveform at the scatterer (output
383         of forward_propagation()).
384     scatterer : dict
385         Must contain "amp" (dimensionless synthetic reflectivity).
386
387     Returns
388     -----
389     np.ndarray
390         Reflected/scattered mechanical pressure waveform, same shape as
391         incident_wave.
392     """
393     return incident_wave * scatterer["amp"]
394
395
396 def return_propagation(
397     reflected_wave: np.ndarray,
398     t_axis: np.ndarray,
399     scatterer: Dict[str, float],
400     element_x: np.ndarray,
401     f0: float = F0,
402     c: float = C,
403     alpha0: float = ALPHA_0,
404 ) -> Tuple[np.ndarray, np.ndarray, np.ndarray]:
405     """
406     Return propagation: scatterer -> every receive element (Part 3).
407
408         d_RX(m) = sqrt((x_s - x_m)^2 + z_s^2)
409         t_RX(m) = d_RX(m) / C
410         p_RX,m(t) = A_RX(m) * p_scattered(t - t_RX(m))
411
412     ``reflected_wave`` is already a discretely SAMPLED array (it came out

```

```

413 of mock_tissue_interaction(), not a closed-form pulse), so shifting it
414 by a continuous delay t_RX(m) is done by linear interpolation onto the
415 shared time axis (np.interp), rather than re-evaluating a formula.
416
417 Parameters
418 -----
419 reflected_wave : np.ndarray, shape (n_samples,)
420     Reflected mechanical pressure waveform at the scatterer, sampled
421     on t_axis (output of mock_tissue_interaction() or, later, a real
422     Tissue Interaction module - this function does not care which).
423 t_axis : np.ndarray, shape (n_samples,)
424     Shared simulation time axis [s].
425 scatterer : dict
426     Must contain "x" and "z" (scatterer position [m]).
427 element_x : array_like, shape (n_elements,)
428     RX element x-positions [m].
429 f0, c, alpha0 : float
430     Physical parameters, default to the global config values.
431
432 Returns
433 -----
434 pressure : np.ndarray, shape (n_elements, n_samples)
435     Returning mechanical acoustic pressure at every receive element.
436 d_rx : np.ndarray, shape (n_elements,)
437     One-way RX distance for each element (diagnostics/plots).
438 t_rx : np.ndarray, shape (n_elements,)
439     One-way RX propagation delay for each element (diagnostics/plots).
440 """
441 element_x = np.asarray(element_x, dtype=float)
442 t_axis = np.asarray(t_axis, dtype=float)
443 reflected_wave = np.asarray(reflected_wave, dtype=float)
444
445 d_rx = calculate_rx_distance(element_x, scatterer["x"], scatterer["z"])
446 t_rx = calculate_propagation_delay(d_rx, c=c)
447 atten_rx = calculate_attenuation(d_rx, f0=f0, alpha0=alpha0)
448
449 n_elements = element_x.size
450 pressure = np.zeros((n_elements, t_axis.size))
451 for m in range(n_elements):
452     # p_scattered(t - t_RX(m)), evaluated on the shared time axis by
453     # interpolating the sampled reflected_wave. Samples that would
454     # fall outside the simulated time window are treated as zero.
455     shifted = np.interp(
456         t_axis - t_rx[m], t_axis, reflected_wave, left=0.0, right=0.0
457     )
458     pressure[m, :] = atten_rx[m] * shifted
459
460 return pressure, d_rx, t_rx
461
462
463 def simulate_acoustic_propagation(
464     phantom_points: List[Dict[str, float]] = None,
465     n_elements: int = N_ELEMENTS,
466     pitch: float = PITCH,
467     f0: float = F0,
468     fs: float = FS,
469     c: float = C,
470     alpha0: float = ALPHA_0,
471     n_cycles: int = N_CYCLES,
472     time_margin_fraction: float = 0.15,
473 ) -> Dict[str, object]:
474     """
475     Run the complete Phase 1 acoustic propagation pipeline:
476
477     TX elements -> forward_propagation -> mock_tissue_interaction
478                 -> return_propagation -> pressure[64, time]
479
480     for every synthetic scatterer in ``phantom_points``, combining
481     multiple scatterers' echoes by linear superposition at each receive
482     element.
483
484     Parameters
485     -----
486     phantom_points : list of dict, optional
487         Synthetic scatterers, each with "x", "z", "amp". Defaults to the
488         global PHANTOM_POINTS.
489     n_elements, pitch, f0, fs, c, alpha0, n_cycles : physical parameters

```

```

490     Default to the global config values.
491     time_margin_fraction : float
492         Extra fraction of the max round-trip time added as head/tail
493         margin on the simulation time axis, so delayed pulses are not
494         clipped.
495
496     Returns
497     -----
498     dict with keys:
499         "t_axis"      : np.ndarray, shape (n_samples,) - shared time axis [s]
500         "pressure"    : np.ndarray, shape (n_elements, n_samples) - FINAL
501                         module output: received mechanical pressure at
502                         every RX element vs time, ALL scatterers summed
503                         (pressure[receive_element, time], per the spec)
504         "received_pressure" : same array as "pressure" (clearer alias)
505         "incident_pressure" : np.ndarray, shape (n_scatterers, n_samples) -
506                         forward_propagation()'s incident wave AT EACH
507                         SCATTERER'S LOCATION vs time, one row per
508                         phantom point (row i <-> phantom_points[i])
509         "reflected_pressure" : np.ndarray, shape (n_scatterers, n_samples) -
510                         mock_tissue_interaction()'s reflected wave AT
511                         EACH SCATTERER'S LOCATION vs time, same row order
512         "element_x"     : np.ndarray, shape (n_elements,) - array geometry
513         "pulse_t"       : np.ndarray - TX pulse time axis (for plotting)
514         "pulse_p"       : np.ndarray - TX pulse waveform (for plotting)
515         "per_scatterer" : list of dict, one per phantom point, each with
516                         "scatterer", "d_tx", "t_tx", "d_rx", "t_rx",
517                         "incident_wave", "reflected_wave" (both at that
518                         scatterer's location vs time), and "pressure"
519                         (that scatterer's own RX contribution, useful
520                         for validating linear superposition)
521
522     """
523     if phantom_points is None:
524         phantom_points = PHANTOM_POINTS
525
526     element_x = create_linear_array(n_elements=n_elements, pitch=pitch)
527     pulse_t, pulse_p = generate_pulse(fs=fs, f0=f0, n_cycles=n_cycles)
528
529     # Size the shared time axis from the largest total (TX+RX) one-way
530     # delay across all scatterers/elements, so no delayed pulse is clipped.
531     max_total_delay = 0.0
532     for scatterer in phantom_points:
533         d_tx = calculate_tx_distance(element_x, scatterer["x"], scatterer["z"])
534         d_rx = calculate_rx_distance(element_x, scatterer["x"], scatterer["z"])
535         t_tx = calculate_propagation_delay(d_tx, c=c)
536         t_rx = calculate_propagation_delay(d_rx, c=c)
537         max_total_delay = max(max_total_delay, float(np.max(t_tx) + np.max(t_rx)))
538
539     pulse_half_width = pulse_t[-1] # generate_pulse() is symmetric about 0
540     margin = time_margin_fraction * max_total_delay + pulse_half_width
541     t_start = -pulse_half_width
542     t_end = max_total_delay + margin
543     t_axis = np.arange(t_start, t_end, 1.0 / fs)
544
545     pressure_total = np.zeros((n_elements, t_axis.size))
546     per_scatterer = []
547     incident_list = [] # one incident-wave row per scatterer (per-LOCATION pressure vs time)
548     reflected_list = [] # one reflected-wave row per scatterer
549
550     for scatterer in phantom_points:
551         incident_wave, d_tx, t_tx = forward_propagation(
552             scatterer, element_x, t_axis, f0=f0, c=c, alpha0=alpha0, n_cycles=n_cycles
553         )
554         reflected_wave = mock_tissue_interaction(incident_wave, scatterer)
555         pressure_s, d_rx, t_rx = return_propagation(
556             reflected_wave, t_axis, scatterer, element_x, f0=f0, c=c, alpha0=alpha0
557         )
558
559         pressure_total += pressure_s # linear superposition across scatterers
560         incident_list.append(incident_wave)
561         reflected_list.append(reflected_wave)
562
563     per_scatterer.append(
564         {
565             "scatterer": scatterer,
566             "d_tx": d_tx,
567             "t_tx": t_tx,

```

```

567         "d_rx": d_rx,
568         "t_rx": t_rx,
569         "incident_wave": incident_wave,    # pressure at THIS scatterer vs time
570         "reflected_wave": reflected_wave, # pressure at THIS scatterer vs time
571         "pressure": pressure_s,            # this scatterer's own RX contribution
572     }
573 )
574
575 return {
576     "t_axis": t_axis,
577     # Received mechanical pressure at all 64 RX elements vs time -- the
578     # module's final output, pressure[receive_element, time].
579     "pressure": pressure_total,
580     "received_pressure": pressure_total, # alias, same array, clearer name
581     # Incident/reflected pressure AT EACH SCATTERER LOCATION vs time,
582     # stacked into one array per stage: shape (n_scatterers, n_time).
583     # Row i corresponds to phantom_points[i] / per_scatterer[i].
584     "incident_pressure": np.stack(incident_list, axis=0),
585     "reflected_pressure": np.stack(reflected_list, axis=0),
586     "element_x": element_x,
587     "pulse_t": pulse_t,
588     "pulse_p": pulse_p,
589     "per_scatterer": per_scatterer,
590 }

```

code/nitk_usoundsim/demo_acoustic_propagation.py

```

1  """
2  demo_acoustic_propagation.py
3  =====
4
5  Small end-to-end demo of the Phase-1 acoustic propagation pipeline:
6
7      64 TX elements -> forward_propagation() -> mock_tissue_interaction()
8                      -> return_propagation() -> pressure[64, time]
9
10 run on the three synthetic PHANTOM_POINTS scatterers from config.py.
11 Saves two PNG figures illustrating every stage of the pipeline.
12
13 Run with (from the usound_sim folder, so the package is importable):
14 python -m nitk_usoundsim.demo_acoustic_propagation
15 """
16
17 import os
18
19 import matplotlib
20 matplotlib.use("Agg")
21 import matplotlib.pyplot as plt
22 import numpy as np
23
24 try:
25     from nitk_usoundsim import config
26     from nitk_usoundsim import acoustic_propagation as ap
27 except ImportError: # allow running directly from inside the package dir
28     import config
29     import acoustic_propagation as ap
30
31 OUT_DIR = os.path.dirname(os.path.abspath(__file__))
32 COLORS = ["tab:blue", "tab:orange", "tab:green"]
33
34
35 def make_figures(out_dir: str = OUT_DIR):
36     result = ap.simulate_acoustic_propagation()
37     x_el = result["element_x"]
38     t_axis = result["t_axis"]
39     pressure = result["pressure"]
40     pulse_t, pulse_p = result["pulse_t"], result["pulse_p"]
41
42     # =====
43     # Figure 1: array geometry, TX pulse, delay-vs-depth, attenuation-vs-depth
44     # =====
45     fig1, axes = plt.subplots(2, 2, figsize=(11, 8))
46
47     # (a) 64-element array geometry + phantom scatterers
48     ax = axes[0, 0]
49     ax.scatter(x_el * 1e3, np.zeros_like(x_el), marker="s", s=18,

```

```

50         color="tab:blue", label="TX/RX elements")
51     for pt, c in zip(config.PHANTOM_POINTS, COLORS):
52         ax.scatter(pt["x"] * 1e3, pt["z"] * 1e3, marker="*", s=180, color=c, zorder=3)
53         ax.annotate(f"z={pt['z']*1e3:.0f} mm", (pt["x"] * 1e3, pt["z"] * 1e3),
54                    textcoords="offset points", xytext=(8, 0), fontsize=9)
55     ax.set_xlabel("Lateral x [mm]")
56     ax.set_ylabel("Depth z [mm]")
57     ax.set_title("64-element linear array + phantom scatterers")
58     ax.invert_yaxis()
59     ax.legend(loc="lower right")
60     ax.grid(alpha=0.3)
61
62     # (b) transmitted pulse
63     ax = axes[0, 1]
64     ax.plot(pulse_t * 1e6, pulse_p, color="tab:blue")
65     ax.set_xlabel("Time [\u00b5s]")
66     ax.set_ylabel("Pressure (a.u.)")
67     ax.set_title(f"Transmitted pulse ({config.F0/1e6:.0f} MHz, "
68                f"{ap.N_CYCLES:.0f}-cycle Gaussian envelope)")
69     ax.grid(alpha=0.3)
70
71     # (c) propagation delay vs depth (on-axis, near-central element)
72     x_center = float(x_el[config.N_ELEMENTS // 2])
73     z_sweep = np.linspace(config.Z_MIN, config.Z_MAX, 200)
74     d_sweep = ap.calculate_tx_distance(x_center, 0.0, z_sweep)
75     t_sweep = ap.calculate_propagation_delay(d_sweep, c=config.C)
76
77     ax = axes[1, 0]
78     ax.plot(z_sweep * 1e3, t_sweep * 1e6, color="tab:green")
79     for pt, c in zip(config.PHANTOM_POINTS, COLORS):
80         ax.axvline(pt["z"] * 1e3, color=c, ls="--", alpha=0.6)
81     ax.set_xlabel("Depth z [mm]")
82     ax.set_ylabel("One-way delay [\u00b5s]")
83     ax.set_title("Propagation delay vs depth (on-axis)")
84     ax.grid(alpha=0.3)
85
86     # (d) attenuation vs depth
87     a_sweep = ap.calculate_attenuation(d_sweep, f0=config.F0, alpha0=config.ALPHA_0)
88     a_sweep_db = -20.0 * np.log10(a_sweep)
89
90     ax = axes[1, 1]
91     ax.plot(z_sweep * 1e3, a_sweep_db, color="tab:purple")
92     for pt, c in zip(config.PHANTOM_POINTS, COLORS):
93         ax.axvline(pt["z"] * 1e3, color=c, ls="--", alpha=0.6)
94     ax.set_xlabel("Depth z [mm]")
95     ax.set_ylabel("One-way attenuation [dB]")
96     ax.set_title(f"Attenuation vs depth (at F0 = {config.F0/1e6:.0f} MHz)")
97     ax.grid(alpha=0.3)
98
99     fig1.tight_layout()
100    fig1_path = os.path.join(out_dir, "demo_geometry_pulse_delay_attenuation.png")
101    fig1.savefig(fig1_path, dpi=140)
102    plt.close(fig1)
103
104    # =====
105    # Figure 2: pipeline signals (incident / reflected / one RX / RX image)
106    # =====
107    fig2, axes2 = plt.subplots(2, 2, figsize=(11, 8))
108
109    # (a) incident wave AT each scatterer (forward_propagation output)
110    ax = axes2[0, 0]
111    incidents = {}
112    for entry, c in zip(result["per_scatterer"], COLORS):
113        scatterer = entry["scatterer"]
114        incident, _, _ = ap.forward_propagation(scatterer, x_el, t_axis)
115        incidents[id(scatterer)] = incident
116        ax.plot(t_axis * 1e6, incident, color=c, label=f"z={scatterer['z']*1e3:.0f} mm")
117    ax.set_xlabel("Time [\u00b5s]")
118    ax.set_ylabel("Incident pressure (a.u.)")
119    ax.set_title("Forward propagation: incident wave at each scatterer")
120    ax.legend()
121    ax.grid(alpha=0.3)
122
123    # (b) synthetic reflected/scattered waves (mock_tissue_interaction output)
124    ax = axes2[0, 1]
125    for entry, c in zip(result["per_scatterer"], COLORS):
126        scatterer = entry["scatterer"]

```

```

127     reflected = ap.mock_tissue_interaction(incidents[id(scatterer)], scatterer)
128     ax.plot(t_axis * 1e6, reflected, color=c,
129             label=f"z={scatterer['z']*1e3:.0f} mm, amp={scatterer['amp']:.1f}")
130     ax.set_xlabel("Time [\u00b5s]")
131     ax.set_ylabel("Reflected pressure (a.u.)")
132     ax.set_title("Synthetic (mock) reflected/scattered waves")
133     ax.legend()
134     ax.grid(alpha=0.3)
135
136     # (c) one receive-element pressure waveform (central element)
137     ax = axes2[1, 0]
138     m_center = config.N_ELEMENTS // 2
139     ax.plot(t_axis * 1e6, pressure[m_center, :], color="black")
140     for pt, c in zip(config.PHANTOM_POINTS, COLORS):
141         expected_rt = 2 * pt["z"] / config.C
142         ax.axvline(expected_rt * 1e6, color=c, ls=":", alpha=0.6)
143     ax.set_xlabel("Time [\u00b5s]")
144     ax.set_ylabel("Pressure (a.u.)")
145     ax.set_title(f"Return propagation: RX element #{m_center} waveform")
146     ax.text(0.02, 0.02, "dotted = expected same-path\nround-trip time per target",
147            transform=ax.transAxes, fontsize=8, va="bottom", ha="left", alpha=0.8)
148     ax.grid(alpha=0.3)
149
150     # (d) receive-element x time image of the full returning pressure field
151     ax = axes2[1, 1]
152     extent = [t_axis[0] * 1e6, t_axis[-1] * 1e6, config.N_ELEMENTS - 1, 0]
153     vmax = np.max(np.abs(pressure)) or 1.0
154     im = ax.imshow(pressure, aspect="auto", extent=extent, cmap="seismic",
155                   vmin=-vmax, vmax=vmax)
156     ax.set_xlabel("Time [\u00b5s]")
157     ax.set_ylabel("RX element index")
158     ax.set_title("Returning mechanical pressure field (all 64 RX elements)")
159     fig2.colorbar(im, ax=ax, label="Pressure (a.u.)")
160
161     fig2.tight_layout()
162     fig2_path = os.path.join(out_dir, "demo_pipeline_signals.png")
163     fig2.savefig(fig2_path, dpi=140)
164     plt.close(fig2)
165
166     return fig1_path, fig2_path, result
167
168
169 def save_part_outputs(result, out_dir: str = OUT_DIR):
170     """
171     Save this module's two deliverables into SEPARATE, self-describing
172     .npz files -- one per project part -- so each can be shown/inspected
173     independently:
174
175     PART 1  forward_propagation()  TX array -> scatterer
176            -> part1_forward_propagation_outputs.npz
177
178     PART 2  return_propagation()   scatterer -> RX array (FINAL output)
179            -> part2_return_propagation_outputs.npz
180
181     Every numeric array is paired with a "*_units" and/or "*_row_labels"
182     entry, and each file has a top-level "description" string, so the
183     contents are understandable on their own when printed -- no need to
184     have this source file open alongside them.
185     """
186     scatterer_labels = np.array([
187         f"scatterer_{i}: x={e['scatterer']['x']*1e3:.1f}mm, "
188         f"z={e['scatterer']['z']*1e3:.1f}mm, amp={e['scatterer']['amp']:.2f}"
189         for i, e in enumerate(result["per_scatterer"])
190     ])
191     scatterer_xyz = np.array(
192         [[e["scatterer"]["x"], e["scatterer"]["z"], e["scatterer"]["amp"]]
193          for e in result["per_scatterer"]]
194     )
195     rx_labels = np.array(
196         [f"rx_element_{m:02d}" for m in range(result["received_pressure"].shape[0])]
197     )
198
199     # ----- PART 1: forward propagation (TX array -> scatterer) ----
200     part1_path = os.path.join(out_dir, "part1_forward_propagation_outputs.npz")
201     np.savez(
202         part1_path,
203         description=np.array(

```

```

204         "PART 1 - forward_propagation(): incident mechanical pressure wave "
205         "arriving AT EACH SCATTERER from the 64-element TX array, vs time. "
206         "Row i of incident_pressure corresponds to scatterer_xyz[i] / "
207         "incident_pressure_row_labels[i]. Also included: the TX pulse "
208         "(pulse_t, pulse_p) and TX element geometry (element_x) used to "
209         "compute it."
210     ),
211     t_axis=result["t_axis"],
212     t_axis_units=np.array("seconds"),
213     incident_pressure=result["incident_pressure"],
214     incident_pressure_shape_meaning=np.array("(n_scatterers, n_time_samples)"),
215     incident_pressure_units=np.array("arbitrary pressure units (a.u.)"),
216     incident_pressure_row_labels=scatterer_labels,
217     scatterer_xyz=scatterer_xyz,
218     scatterer_xyz_columns=np.array(["x [m]", "z [m]", "amp"]),
219     element_x=result["element_x"],
220     element_x_units=np.array("meters (TX element lateral position)"),
221     pulse_t=result["pulse_t"],
222     pulse_p=result["pulse_p"],
223 )
224
225 # ----- PART 2: return propagation (scatterer -> RX array) -----
226 part2_path = os.path.join(out_dir, "part2_return_propagation_outputs.npz")
227 np.savez(
228     part2_path,
229     description=np.array(
230         "PART 2 - return_propagation(): FINAL module output. Mechanical "
231         "pressure received at each of the 64 RX elements vs time "
232         "(received_pressure), all scatterers combined by linear "
233         "superposition. reflected_pressure_input is the reflected/"
234         "scattered wave this stage propagates to the RX array -- it is "
235         "INCLUDED HERE ONLY FOR CONTEXT (currently produced by the "
236         "temporary mock tissue-interaction stand-in; return_propagation() "
237         "itself does not care how it was produced, so the real tissue-"
238         "interaction module's output can replace it later unchanged).",
239     ),
240     t_axis=result["t_axis"],
241     t_axis_units=np.array("seconds"),
242     received_pressure=result["received_pressure"],
243     received_pressure_shape_meaning=np.array("(64 rx_elements, n_time_samples)"),
244     received_pressure_units=np.array("arbitrary pressure units (a.u.)"),
245     received_pressure_row_labels=rx_labels,
246     reflected_pressure_input=result["reflected_pressure"],
247     reflected_pressure_input_row_labels=scatterer_labels,
248     scatterer_xyz=scatterer_xyz,
249     scatterer_xyz_columns=np.array(["x [m]", "z [m]", "amp"]),
250     element_x=result["element_x"],
251     element_x_units=np.array("meters (RX element lateral position)"),
252 )
253
254 return part1_path, part2_path
255
256
257 if __name__ == "__main__":
258     f1, f2, result = make_figures()
259     part1_path, part2_path = save_part_outputs(result)
260
261     print("Saved figure:", f1)
262     print("Saved figure:", f2)
263     print("Saved PART 1 (forward propagation) data:", part1_path)
264     print("Saved PART 2 (return propagation) data: ", part2_path)
265     print()
266     print("PART 1 - incident_pressure :", result["incident_pressure"].shape,
267           " <- pressure AT each scatterer, per phantom point")
268     print("PART 2 - received_pressure :", result["received_pressure"].shape,
269           " <- FINAL output, pressure[rx_element, time]")
270     print()
271     print("To load and inspect later, e.g.:")
272     print(" import numpy as np")
273     print(" d1 = np.load('part1_forward_propagation_outputs.npz')")
274     print(" print(d1['description'])")
275     print(" print(d1.files)                                # lists every labelled array")
276     print(" d1['incident_pressure']                          # (n_scatterers, n_time)")
277     print(" d1['incident_pressure_row_labels']                  # which row is which scatterer")
278     print()
279     print(" d2 = np.load('part2_return_propagation_outputs.npz')")
280     print(" print(d2['description'])")

```

```
281     print("    d2['received_pressure']                # (64, n_time)")
```

code/nitk_usoundsim/test_acoustic_propagation.py

```
1  """
2  test_acoustic_propagation.py
3  =====
4
5  Basic unit tests for the Phase-1 Acoustic Propagation module
6  (nitk_usoundsim.acoustic_propagation), covering:
7
8  - element positions
9  - wavelength and pitch
10 - TX distance calculation
11 - RX distance calculation
12 - TX propagation delay
13 - RX propagation delay
14 - attenuation calculation
15 - output dimensions
16 - expected round-trip delay for the 20 mm phantom target
17 - deeper targets -> larger delay / more attenuation (monotonicity)
18 - off-axis targets -> different RX delays across elements
19 - multiple-scatterer linear superposition
20
21 Run with: python -m unittest nitk_usoundsim.test_acoustic_propagation -v
22 (from the usound_sim folder, so the `nitk_usoundsim` package is importable).
23 """
24
25 import unittest
26
27 import numpy as np
28
29 from nitk_usoundsim import config
30 from nitk_usoundsim import acoustic_propagation as ap
31
32
33 class TestArrayGeometry(unittest.TestCase):
34     def test_element_count(self):
35         x = ap.create_linear_array()
36         self.assertEqual(x.size, config.N_ELEMENTS)
37
38     def test_element_spacing_equals_pitch(self):
39         x = ap.create_linear_array()
40         spacing = np.diff(x)
41         np.testing.assert_allclose(spacing, config.PITCH, rtol=1e-12)
42
43     def test_array_centred_on_zero(self):
44         x = ap.create_linear_array()
45         self.assertAlmostEqual(float(np.mean(x)), 0.0, places=12)
46
47     def test_wavelength_and_pitch_relationship(self):
48         # LAMBDA = C / F0 ; PITCH = LAMBDA / 2 (both come from config.py)
49         expected_lambda = config.C / config.F0
50         self.assertAlmostEqual(config.LAMBDA, expected_lambda, places=9)
51         self.assertAlmostEqual(config.PITCH, expected_lambda / 2.0, places=9)
52
53
54 class TestDistancesAndDelays(unittest.TestCase):
55     def setUp(self):
56         self.x_elements = ap.create_linear_array()
57
58     def test_tx_distance_matches_manual_formula(self):
59         x_s, z_s = 0.003, 0.020
60         d = ap.calculate_tx_distance(self.x_elements, x_s, z_s)
61         expected = np.sqrt((x_s - self.x_elements) ** 2 + z_s ** 2)
62         np.testing.assert_allclose(d, expected, rtol=1e-12)
63
64     def test_rx_distance_matches_manual_formula(self):
65         x_s, z_s = -0.004, 0.030
66         d = ap.calculate_rx_distance(self.x_elements, x_s, z_s)
67         expected = np.sqrt((x_s - self.x_elements) ** 2 + z_s ** 2)
68         np.testing.assert_allclose(d, expected, rtol=1e-12)
69
70     def test_tx_distance_on_axis_center_equals_depth(self):
71         # For a scatterer directly on axis (x_s=0), the two central
72         # elements should be almost exactly `z_s` away.
```

```

73     d = ap.calculate_tx_distance(self.x_elements, 0.0, 0.020)
74     self.assertAlmostEqual(d.min(), 0.020, delta=1e-4)
75
76     def test_tx_propagation_delay(self):
77         d = np.array([0.010, 0.020, 0.030])
78         t = ap.calculate_propagation_delay(d, c=config.C)
79         np.testing.assert_allclose(t, d / config.C, rtol=1e-12)
80
81     def test_rx_propagation_delay(self):
82         d = np.array([0.015, 0.025])
83         t = ap.calculate_propagation_delay(d, c=config.C)
84         np.testing.assert_allclose(t, d / config.C, rtol=1e-12)
85
86     def test_deeper_targets_have_larger_tx_delay(self):
87         d_shallow = ap.calculate_tx_distance(self.x_elements, 0.0, 0.010)
88         d_deep = ap.calculate_tx_distance(self.x_elements, 0.0, 0.030)
89         t_shallow = ap.calculate_propagation_delay(d_shallow, c=config.C)
90         t_deep = ap.calculate_propagation_delay(d_deep, c=config.C)
91         self.assertTrue(np.all(t_deep > t_shallow))
92
93     def test_offaxis_target_gives_varying_rx_delays_across_elements(self):
94         d = ap.calculate_rx_distance(self.x_elements, 0.005, 0.020)
95         t = ap.calculate_propagation_delay(d, c=config.C)
96         # An off-axis target must not arrive at every element at the same time.
97         self.assertGreater(t.max() - t.min(), 0.0)
98
99
100 class TestAttenuation(unittest.TestCase):
101     def test_attenuation_matches_manual_db_formula(self):
102         distance_m = 0.020
103         atten = ap.calculate_attenuation(distance_m, f0=config.F0, alpha0=config.ALPHA_0)
104         f_mhz = config.F0 / 1.0e6
105         d_cm = distance_m * 100.0
106         expected_db = config.ALPHA_0 * f_mhz * d_cm
107         expected_linear = 10.0 ** (-expected_db / 20.0)
108         self.assertAlmostEqual(atten, expected_linear, places=12)
109
110     def test_attenuation_at_20mm_is_5dB(self):
111         # ALPHA_0=0.5 dB/MHz/cm * F0=5MHz * 2cm = 5 dB one-way.
112         atten = ap.calculate_attenuation(0.020, f0=config.F0, alpha0=config.ALPHA_0)
113         atten_db = -20.0 * np.log10(atten)
114         self.assertAlmostEqual(atten_db, 5.0, places=6)
115
116     def test_deeper_targets_experience_more_attenuation(self):
117         atten_shallow = ap.calculate_attenuation(0.010)
118         atten_deep = ap.calculate_attenuation(0.030)
119         # Less remaining amplitude (smaller linear factor) means MORE attenuation.
120         self.assertLess(atten_deep, atten_shallow)
121
122     def test_attenuation_is_bounded_between_0_and_1(self):
123         atten = ap.calculate_attenuation(np.array([0.0, 0.01, 0.02, 0.05]))
124         self.assertTrue(np.all(atten <= 1.0))
125         self.assertTrue(np.all(atten > 0.0))
126
127
128 class TestPulse(unittest.TestCase):
129     def test_pulse_sampled_at_fs(self):
130         t, p = ap.generate_pulse()
131         dt = np.diff(t)
132         np.testing.assert_allclose(dt, 1.0 / config.FS, rtol=1e-9)
133         self.assertEqual(t.shape, p.shape)
134
135     def test_pulse_centered_near_zero(self):
136         t, p = ap.generate_pulse()
137         self.assertAlmostEqual(t[np.argmax(np.abs(p))], 0.0, delta=2.0 / config.FS)
138
139
140 class TestFullPipeline(unittest.TestCase):
141     @classmethod
142     def setUpClass(cls):
143         cls.result = ap.simulate_acoustic_propagation()
144
145     def test_output_dimensions(self):
146         pressure = self.result["pressure"]
147         self.assertEqual(pressure.shape[0], config.N_ELEMENTS)
148         self.assertEqual(pressure.shape[1], self.result["t_axis"].size)
149         self.assertGreater(pressure.shape[1], 0)

```

```

150
151     def test_output_is_real_valued_mechanical_pressure(self):
152         pressure = self.result["pressure"]
153         self.assertTrue(np.isrealobj(pressure))
154         self.assertTrue(np.all(np.isfinite(pressure)))
155
156     def test_round_trip_delay_for_20mm_phantom_point(self):
157         # Nominal on-axis, same-path round trip: 2 * 20e-3 / 1540 ~= 25.97 us
158         entry_20mm = next(
159             e for e in self.result["per_scatterer"] if abs(e["scatterer"]["z"] - 0.020) < 1e-9
160         )
161         round_trip_min = entry_20mm["t_tx"].min() + entry_20mm["t_rx"].min()
162         expected = 2.0 * 0.020 / config.C
163         self.assertAlmostEqual(round_trip_min, expected, delta=0.05e-6) # 50 ns tolerance
164
165     def test_deeper_phantom_points_have_larger_min_delay(self):
166         entries = sorted(self.result["per_scatterer"], key=lambda e: e["scatterer"]["z"])
167         min_delays = [e["t_tx"].min() for e in entries]
168         self.assertTrue(np.all(np.diff(min_delays) > 0))
169
170     def test_multiple_scatterer_linear_superposition(self):
171         total = self.result["pressure"]
172         manual_sum = np.zeros_like(total)
173         for entry in self.result["per_scatterer"]:
174             manual_sum += entry["pressure"]
175         np.testing.assert_allclose(total, manual_sum, rtol=1e-12, atol=1e-15)
176
177
178 if __name__ == "__main__":
179     unittest.main(verbosity=2)

```

code/nitk_usoundsim/check_arr.py

```

1 import numpy as np
2 #data = np.load("nitk_usoundsim/acoustic_propagation_outputs.npz")
3 #print(data["received_pressure"])
4
5 '''data = np.load("nitk_usoundsim/part1_forward_propagation_outputs.npz")
6 print(data["description"])
7 print(data.files)'''
8
9
10 data = np.load("nitk_usoundsim/part2_return_propagation_outputs.npz")
11 print(data["description"])
12 print(data["reflected_pressure_input"])
13 print(data["reflected_pressure_input"].shape)
14 print(data["reflected_pressure_input_row_labels"])

```

code/nitk_usoundsim/requirements.txt

```

1 numpy
2 matplotlib

```

run_team3.py (runner added for this demo)

```

1 """
2 run_team3.py - Team 3: Acoustic Propagation (NITK-UsoundSim)
3 =====
4
5 ONE command runs the whole Team 3 demo:
6
7     python3 run_team3.py      (macOS / Linux)
8     python  run_team3.py      (Windows)
9
10 What it does (the team's code in code/nitk_usoundsim/ is NOT changed;
11 this script only imports it and calls its functions):
12
13     STEP 1 Run the team's 22 unit tests (test_acoustic_propagation.py).
14     STEP 2 Run the team's own demo (64 elements, 5 MHz, 3 scatterers)
15            and save its two figures + two .npz data files into output/.
16     STEP 3 Second run with the project's final probe (Philips L12-4 model:
17            128 elements, 0.30 mm pitch, 8 MHz) and one point scatterer at
18            (0, 20) mm, done only by passing arguments to the team's functions.

```

```

19     STEP 4 Attenuation vs depth at 4, 8 and 12 MHz (team's calculate_attenuation).
20
21 Everything is saved in output/ (plots, data, console log, console screenshot).
22 """
23
24 import io
25 import json
26 import os
27 import subprocess
28 import sys
29 import time
30
31 sys.dont_write_bytecode = True          # keep code/ clean (no __pycache__)
32
33 HERE = os.path.dirname(os.path.abspath(__file__))
34 CODE_DIR = os.path.join(HERE, "code")
35 INPUT_FILE = os.path.join(HERE, "input", "team3_run_parameters.json")
36 OUT_DIR = os.path.join(HERE, "output")
37 sys.path.insert(0, CODE_DIR)           # makes the team's package importable
38
39 import matplotlib                      # noqa: E402
40 matplotlib.use("Agg")
41 import matplotlib.pyplot as plt        # noqa: E402
42 import numpy as np                    # noqa: E402
43
44 from nitk_usoundsim import config      # noqa: E402 (team code)
45 from nitk_usoundsim import acoustic_propagation as ap    # noqa: E402 (team code)
46 from nitk_usoundsim import demo_acoustic_propagation as demo # noqa: E402 (team code)
47
48 # -----
49 # Small helpers (printing to screen AND to a log file)
50 # -----
51
52 class Tee:
53     """Send printed text to the real console and keep a copy for the log."""
54
55     def __init__(self, stream):
56         self.stream = stream
57         self.buffer = io.StringIO()
58
59     def write(self, text):
60         self.stream.write(text)
61         self.buffer.write(text)
62
63     def flush(self):
64         self.stream.flush()
65
66
67 def banner(title):
68     print()
69     print("=" * 74)
70     print(title)
71     print("=" * 74)
72
73
74 def rel(path):
75     return os.path.relpath(path, HERE)
76
77
78 def console_to_png(text, out_png, title="Terminal", width_chars=100, max_lines=150):
79     """Draw the captured console text as a terminal-style screenshot PNG."""
80     lines = []
81     for line in text.rstrip("\n").expandtabs(4).splitlines() or [""]:
82         while len(line) > width_chars:
83             lines.append(line[:width_chars])
84             line = " -> " + line[width_chars:]
85         lines.append(line)
86     if len(lines) > max_lines:
87         lines = lines[:max_lines - 1] + [f"... ({len(lines) - max_lines + 1} more lines, see console_log.txt)"]
88     h = 0.62 + 0.165 * len(lines)
89     fig = plt.figure(figsize=(0.083 * width_chars + 0.4, h))
90     fig.patch.set_facecolor("#lelele")
91     fig.text(0.012, 1 - 0.28 / h, "●●● " + title, color="#9a9a9a",
92            fontsize=9, family="DejaVu Sans Mono", va="center")
93     for i, ln in enumerate(lines):
94         fig.text(0.012, 1 - (0.55 + 0.165 * i) / h, ln, color="#e6e6e6",

```

```

95         fontsize=8.6, family="DejaVu Sans Mono", va="top")
96     fig.savefig(out_png, dpi=160, facecolor=fig.get_facecolor())
97     plt.close(fig)
98     return out_png
99
100
101 # -----
102 # STEP 1: the team's unit tests
103 # -----
104 def step1_unit_tests():
105     banner("STEP 1: Team 3 unit tests (test_acoustic_propagation.py)")
106     print("Command used: python -m unittest nitk_usoundsim.test_acoustic_propagation -v")
107     print("(run inside the code/ folder)\n")
108     env = dict(os.environ, PYTHONDONTWRITEBYTECODE="1")
109     proc = subprocess.run(
110         [sys.executable, "-m", "unittest", "nitk_usoundsim.test_acoustic_propagation", "-v"],
111         cwd=CODE_DIR, env=env, capture_output=True, text=True)
112     text = (proc.stdout + proc.stderr).strip()
113     with open(os.path.join(OUT_DIR, "unit_tests_full_output.txt"), "w") as fh:
114         fh.write(text + "\n")
115     # Print a short line per test: "ok Class.test_name" (full text is saved)
116     for line in text.splitlines():
117         if " ... " in line and "(" in line:
118             name, _, verdict = line.partition(" ... ")
119             full = name[name.find("(") + 1:name.rfind(")")]
120             short = ".".join(full.split(".")[1:-1])
121             print(f" {verdict.strip():5s} {short}")
122         elif line.startswith("Ran ") or line.strip() == "OK" or line.startswith("FAILED"):
123             print(line)
124     print(" (full unittest text saved in output/unit_tests_full_output.txt)")
125     n_ran = 0
126     for line in text.splitlines():
127         if line.startswith("Ran ") and " test" in line:
128             n_ran = int(line.split()[1])
129     n_ok = sum(1 for line in text.splitlines() if line.rstrip().endswith("... ok"))
130     status = "ALL PASSED" if proc.returncode == 0 else "SOME TESTS FAILED"
131     print(f"\nRESULT: {n_ok} of {n_ran} tests passed -> {status}")
132     return n_ok, n_ran
133
134
135 # -----
136 # STEP 2: the team's own demo (64 elements, 5 MHz, 3 scatterers)
137 # -----
138 def step2_team_demo(params):
139     banner("STEP 2: Team demo - 64 elements, 5 MHz, lambda/2 pitch (config.py)")
140     p1 = params["run1_team_design"]
141     same = (config.N_ELEMENTS == p1["n_elements"] and config.F0 == p1["f0_hz"]
142            and config.FS == p1["fs_hz"] and config.C == p1["c_m_per_s"]
143            and config.ALPHA_0 == p1["alpha0_db_per_mhz_per_cm"]
144            and config.PHANTOM_POINTS == p1["scatterers"])
145     print(f"Input check: values in input JSON match config.py -> {same}")
146     print(f" Elements      : {config.N_ELEMENTS}")
147     print(f" Centre freq    : {config.F0 / 1e6:.1f} MHz")
148     print(f" Sampling freq  : {config.FS / 1e6:.1f} MHz")
149     print(f" Speed of sound: {config.C:.0f} m/s")
150     print(f" Wavelength    : {config.LAMBDA * 1e3:.4f} mm")
151     print(f" Pitch (lam/2) : {config.PITCH * 1e3:.4f} mm")
152     print(f" Attenuation    : {config.ALPHA_0} dB/MHz/cm")
153     print(f" Scatterers     : " + ", ".join(
154         f"({p['x']} * 1e3:.0f), {p['z']} * 1e3:.0f} mm" for p in config.PHANTOM_POINTS))
155
156     # The team's demo normally saves next to its own source file. Here we pass
157     # out_dir=output/ to its functions so the original folder stays untouched.
158     f1, f2, result = demo.make_figures(out_dir=OUT_DIR)
159     part1, part2 = demo.save_part_outputs(result, out_dir=OUT_DIR)
160     print()
161     print("Saved figure:", rel(f1))
162     print("Saved figure:", rel(f2))
163     print("Saved PART 1 (forward propagation) data:", rel(part1))
164     print("Saved PART 2 (return propagation) data: ", rel(part2))
165
166     t = result["t_axis"]
167     print()
168     print(f"Time axis      : {t.size} samples, {t[0] * 1e6:.3f} to {t[-1] * 1e6:.3f} us")
169     print(f"PART 1 incident_pressure shape : {result['incident_pressure'].shape}")
170     print(f"      (scatterers, time)")
171     print(f"PART 2 received_pressure shape : {result['received_pressure'].shape}")

```

```

172     " (RX elements, time)")
173     print("Per scatterer (on-axis, round trip = min t_TX + min t_RX):")
174     for e in result["per_scatterer"]:
175         s = e["scatterer"]
176         rt = (e["t_tx"].min() + e["t_rx"].min()) * 1e6
177         a = ap.calculate_attenuation(e["d_tx"].min()) ** 2
178         print(f"  z = {s['z'] * 1e3:2.0f} mm: one-way t = {e['t_tx'].min() * 1e6:6.3f}-"
179               f"{e['t_tx'].max() * 1e6:6.3f} us | round trip = {rt:6.3f} us "
180               f"{(2z/c = {2 * s['z'] / config.C * 1e6:6.3f}) | 2-way amp = {a:.3f}")
181     return result
182
183
184 # -----
185 # STEP 3: project's final probe (Philips L12-4 model), by arguments only
186 # -----
187 def step3_final_probe(params):
188     p = params["run2_final_probe"]
189     banner("STEP 3: Final probe - Philips L12-4 model (128 el, 0.30 mm, 8 MHz)")
190     print("Done ONLY by passing arguments to the team's functions (code unchanged).")
191     n_el, pitch, f0, fs = p["n_elements"], p["pitch_m"], p["f0_hz"], p["fs_hz"]
192     scat = dict(p["scatterer"])
193
194     element_x = ap.create_linear_array(n_el, pitch)
195     pulse_t, pulse_p = ap.generate_pulse(fs=fs, f0=f0)
196     half_width = pulse_t[-1]
197
198     # Build the shared time axis the same way the team's simulator does:
199     # start at minus the pulse half-width, end after the longest round trip.
200     t_tx_all = ap.calculate_propagation_delay(
201         ap.calculate_tx_distance(element_x, scat["x"], scat["z"]))
202     t_rx_all = ap.calculate_propagation_delay(
203         ap.calculate_rx_distance(element_x, scat["x"], scat["z"]))
204     max_total = float(t_tx_all.max() + t_rx_all.max())
205     t_end = max_total + p["time_margin_fraction"] * max_total + half_width
206     t_axis = np.arange(-half_width, t_end, 1.0 / fs)
207
208     incident, d_tx, t_tx = ap.forward_propagation(scat, element_x, t_axis, f0=f0)
209     reflected = ap.mock_tissue_interaction(incident, scat)
210     pressure, d_rx, t_rx = ap.return_propagation(reflected, t_axis, scat, element_x, f0=f0)
211
212     print(f" Elements      : {n_el}    pitch = {pitch * 1e3:.2f} mm "
213           f"aperture = {(element_x[-1] - element_x[0]) * 1e3:.2f} mm")
214     print(f" Element x range : {element_x[0] * 1e3:.2f} to {element_x[-1] * 1e3:.2f} mm")
215     print(f" Pulse          : {f0 / 1e6:.0f} MHz, {ap.N_CYCLES} cycles, sigma = "
216           f"{ap.N_CYCLES / (2 * f0) * 1e6:.4f} us, half-width = {half_width * 1e6:.4f} us, "
217           f"{pulse_t.size} samples")
218     print(f" Scatterer       : ({scat['x'] * 1e3:.0f}, {scat['z'] * 1e3:.0f}) mm, amp = {scat['am"
219           "p"]}")
220     print(f" Time axis      : {t_axis.size} samples at {fs / 1e6:.0f} MHz, "
221           f"{t_axis[0] * 1e6:.3f} to {t_axis[-1] * 1e6:.3f} us")
222     print(f" One-way distance: {d_tx.min() * 1e3:.3f} to {d_tx.max() * 1e3:.3f} mm")
223     print(f" One-way time   : {t_tx.min() * 1e6:.2f} to {t_tx.max() * 1e6:.2f} us "
224           f"(centre to edge elements)")
225     print(f" Round trip time : {(t_tx.min() + t_rx.min()) * 1e6:.2f} us (centre element) to "
226           f"{(t_tx.max() + t_rx.max()) * 1e6:.2f} us (edge element)")
227     a = ap.calculate_attenuation(d_tx, f0=f0)
228     print(f" One-way amplitude factor at 8 MHz: {a.max():.4f} (centre) to {a.min():.4f} (edge)"
229           f" = {-20 * np.log10(a.max()):.2f} to {-20 * np.log10(a.min()):.2f} dB")
230     i_pk = int(np.argmax(np.abs(incident)))
231     print(f" Incident wave peak |p| = {abs(incident[i_pk]):.3f} a.u. at t = {t_axis[i_pk] * 1e6:.
232           "2f} us")
233     f" (sum of {n_el} attenuated pulses)")
234     print(f" Received pressure array shape: {pressure.shape} (RX elements, time)")
235     m_c = n_el // 2
236     j_pk = int(np.argmax(np.abs(pressure[m_c])))
237     print(f" Element #{m_c}: peak |p| = {abs(pressure[m_c, j_pk]):.3f} a.u. at t = {t_axis[j_pk]
238           " * 1e6:.2f} us")
239     print(f" Element #0 : peak |p| = {np.abs(pressure[0]).max():.3f} a.u. at t = "
240           f"{t_axis[np.argmax(np.abs(pressure[0]))] * 1e6:.2f} us")
241
242     # ----- figure -----
243     fig, axs = plt.subplots(2, 2, figsize=(12, 8.5))
244     ax = axs[0, 0]
245     ax.plot(pulse_t * 1e6, pulse_p, color="tab:blue")
246     ax.set_title(f"(a) Transmitted pulse: {f0 / 1e6:.0f} MHz, {ap.N_CYCLES}-cycle Gaussian")
247     ax.set_xlabel("Time [μs]"); ax.set_ylabel("Pressure (a.u.)"); ax.grid(alpha=0.3)

```

```

246 ax = axs[0, 1]
247 ax.plot(t_axis * 1e6, incident, color="tab:orange", lw=0.9)
248 ax.axvspan(t_tx.min() * 1e6, t_tx.max() * 1e6, color="grey", alpha=0.15,
249            label="range of one-way travel times")
250 ax.set_xlim((t_tx.min() - 1.5e-6) * 1e6, (t_tx.max() + 1.5e-6) * 1e6)
251 ax.set_title("(b) Incident wave at the scatterer (0, 20) mm")
252 ax.set_xlabel("Time [μs]"); ax.set_ylabel("Pressure (a.u.)")
253 ax.legend(loc="upper right", fontsize=8); ax.grid(alpha=0.3)
254
255 ax = axs[1, 0]
256 idx = np.arange(n_el)
257 ax.plot(idx, t_tx * 1e6, color="tab:green", label="t_TX (element -> scatterer)")
258 ax.plot(idx, t_rx * 1e6, "k:", label="t_RX (scatterer -> element)")
259 ax.set_title("(c) One-way travel time vs element")
260 ax.set_xlabel("Element index"); ax.set_ylabel("Time [μs]")
261 ax.legend(fontsize=8); ax.grid(alpha=0.3)
262
263 ax = axs[1, 1]
264 vmax = np.max(np.abs(pressure)) or 1.0
265 im = ax.imshow(pressure, aspect="auto", cmap="seismic", vmin=-vmax, vmax=vmax,
266               extent=[t_axis[0] * 1e6, t_axis[-1] * 1e6, n_el - 1, 0])
267 ax.set_xlim(24, 38)
268 ax.set_title(f"(d) Echo received on all {n_el} elements")
269 ax.set_xlabel("Time [μs]"); ax.set_ylabel("RX element index")
270 fig.colorbar(im, ax=ax, label="Pressure (a.u.)")
271 fig.suptitle("Final probe run: Philips L12-4 model, point scatterer at (0, 20) mm", fontweight
272             ← "bold")
273 fig.tight_layout()
274 fig_path = os.path.join(OUT_DIR, "L12-4_final_probe_propagation.png")
275 fig.savefig(fig_path, dpi=140)
276 plt.close(fig)
277
278 npz_path = os.path.join(OUT_DIR, "L12-4_final_probe_outputs.npz")
279 np.savez(npz_path, t_axis=t_axis, element_x=element_x, pulse_t=pulse_t, pulse_p=pulse_p,
280          incident_wave=incident, reflected_wave=reflected, received_pressure=pressure,
281          d_tx=d_tx, t_tx=t_tx, d_rx=d_rx, t_rx=t_rx,
282          description=np.array("Final probe run (128 el, 0.30 mm pitch, 8 MHz), scatterer (0,20
283          ← ) mm. "
284          ← "received_pressure shape = (128 RX elements, time). Units: m, s,
285          ← a.u.))
286 print()
287 print("Saved figure:", rel(fig_path))
288 print("Saved data :", rel(npz_path))
289
290 # -----
291 # STEP 4: attenuation at 4 / 8 / 12 MHz
292 # -----
293 def step4_attenuation(params):
294     p = params["attenuation_plot"]
295     banner("STEP 4: Attenuation vs depth at 4, 8, 12 MHz (calculate_attenuation)")
296     depth = np.linspace(0.0, p["max_depth_m"], 301)
297     fig, axs = plt.subplots(1, 2, figsize=(11, 4))
298     print(f"alpha0 = {config.ALPHA_0} dB/MHz/cm. One-way and two-way (round trip) loss:")
299     print(" f0          depth 20 mm: one-way dB  amp factor | round-trip dB  amp factor")
300     for f in p["frequencies_hz"]:
301         a = ap.calculate_attenuation(depth, f0=f)
302         axs[0].plot(depth * 1e3, -20 * np.log10(a), label=f"{f / 1e6:.0f} MHz")
303         axs[1].plot(depth * 1e3, a ** 2, label=f"{f / 1e6:.0f} MHz")
304         a20 = float(ap.calculate_attenuation(0.020, f0=f))
305         print(f"    {f / 1e6:4.0f} MHz          {-20 * np.log10(a20):6.2f}      {a20:.4f}   | "
306               f"    {-40 * np.log10(a20):6.2f}      {a20 ** 2:.4f}")
307     axs[0].set_title("One-way attenuation (dB)")
308     axs[0].set_xlabel("Depth [mm]"); axs[0].set_ylabel("Loss [dB]")
309     axs[1].set_title("Round-trip amplitude factor (linear, = A_TX x A_RX)")
310     axs[1].set_xlabel("Depth [mm]"); axs[1].set_ylabel("Amplitude factor")
311     for ax in axs:
312         ax.axvline(20, color="grey", ls="--", alpha=0.6)
313         ax.legend(); ax.grid(alpha=0.3)
314     fig.tight_layout()
315     path = os.path.join(OUT_DIR, "attenuation_vs_depth_4_8_12MHz.png")
316     fig.savefig(path, dpi=140)
317     plt.close(fig)
318     print("Saved figure:", rel(path))
319
320 def main():

```

```

320     t0 = time.time()
321     os.makedirs(OUT_DIR, exist_ok=True)
322     tee = Tee(sys.stdout)
323     sys.stdout = tee
324     try:
325         print("NITK-UsoundSim - Team 3: Acoustic Propagation - demo run")
326         print(f"Python {sys.version.split()[0]}, numpy {np.__version__}, matplotlib {matplotlib.__
↵ version__}")
327         print(f"Input file : {rel(INPUT_FILE)}")
328         print(f"Output dir : {rel(OUT_DIR)}/")
329         with open(INPUT_FILE) as fh:
330             params = json.load(fh)
331             n_ok, n_ran = step1_unit_tests()
332             step2_team_demo(params)
333             step3_final_probe(params)
334             step4_attenuation(params)
335             banner("SUMMARY")
336             print(f"Unit tests : {n_ok}/{n_ran} passed")
337             print("Output files in output/:")
338             for name in sorted(os.listdir(OUT_DIR)):
339                 if name not in ("console_log.txt", "console_screenshot.png", "unit_tests_full_output.t
↵ xt"):
340                     print("  - ", name)
341                     print("  - unit_tests_full_output.txt")
342                     print("  - console_log.txt")
343                     print("  - console_screenshot.png")
344                     print(f"Total run time: {time.time() - t0:.1f} s")
345         finally:
346             sys.stdout = tee.stream
347             log = tee.buffer.getvalue()
348             with open(os.path.join(OUT_DIR, "console_log.txt"), "w") as fh:
349                 fh.write(log)
350             console_to_png(log, os.path.join(OUT_DIR, "console_screenshot.png"),
351                               title="python3 run_team3.py (Team 3: Acoustic Propagation)", width_chars=106)
352             print("Console log and screenshot saved in output/. Done.")
353
354
355 if __name__ == "__main__":
356     main()

```

4 Input

What goes into the run

The team's functions take numbers as arguments (not image files), so the input is a small JSON file: `input/team3_run_parameters.json`. Run 1 uses the team's own `config.py`; the JSON only lists those values and `run_team3.py` checks that they match (the run printed `True`). Run 2 and the attenuation plot take their values from the JSON.

Setting	Run 1: team design	Run 2: final probe (L12-4 model)
Elements	64	128
Pitch	$\lambda/2 = 0.154$ mm	0.30 mm
Centre frequency	5 MHz	8 MHz
Sampling frequency	40 MHz	40 MHz
Speed of sound, attenuation	1540 m/s, 0.5 dB/MHz/cm	same (config.py defaults)
Pulse	3-cycle Gaussian	3-cycle Gaussian
Scatterers (x, z), amp	(0, 10), (0, 20), (0, 30) mm, amp 1.0	(0, 20) mm, amp 1.0
Time axis	built by <code>simulate_acoustic_propagation()</code>	built by <code>run_team3.py</code> : start at -half-width, 15 % margin

Attenuation plot: frequencies 4, 8 and 12 MHz, depth 0 to 60 mm.

`input/team3_run_parameters.json`

```

1 {
2   "about": "Sample input for run_team3.py (Team 3: Acoustic Propagation). Distances in metres, fre
   ↳ quencies in Hz, speed in m/s.",
3   "run1_team_design": {
4     "about": "The team's own design. These values are read from code/nitk_usoundsim/config.py; the
   ↳ y are listed here only so you can see them. run_team3.py checks that they match config.py.",
5     "n_elements": 64,
6     "f0_hz": 5000000.0,
7     "fs_hz": 40000000.0,
8     "c_m_per_s": 1540.0,
9     "alpha0_db_per_mhz_per_cm": 0.5,
10    "pitch": "lambda/2 = (1540 / 5e6) / 2 = 0.154 mm",
11    "scatterers": [
12      {"x": 0.000, "z": 0.010, "amp": 1.0},
13      {"x": 0.000, "z": 0.020, "amp": 1.0},
14      {"x": 0.000, "z": 0.030, "amp": 1.0}
15    ]
16  },
17  "run2_final_probe": {
18    "about": "The project's final probe (Philips L12-4 model). These values are passed as argument
   ↳ s to the team's functions; the team's code is not changed.",
19    "probe_name": "Philips L12-4 (project model)",
20    "n_elements": 128,
21    "pitch_m": 0.0003,
22    "f0_hz": 8000000.0,
23    "fs_hz": 40000000.0,
24    "scatterer": {"x": 0.000, "z": 0.020, "amp": 1.0},
25    "time_margin_fraction": 0.15
26  },
27  "attenuation_plot": {
28    "frequencies_hz": [4000000.0, 8000000.0, 12000000.0],
29    "max_depth_m": 0.06
30  }
31 }
```

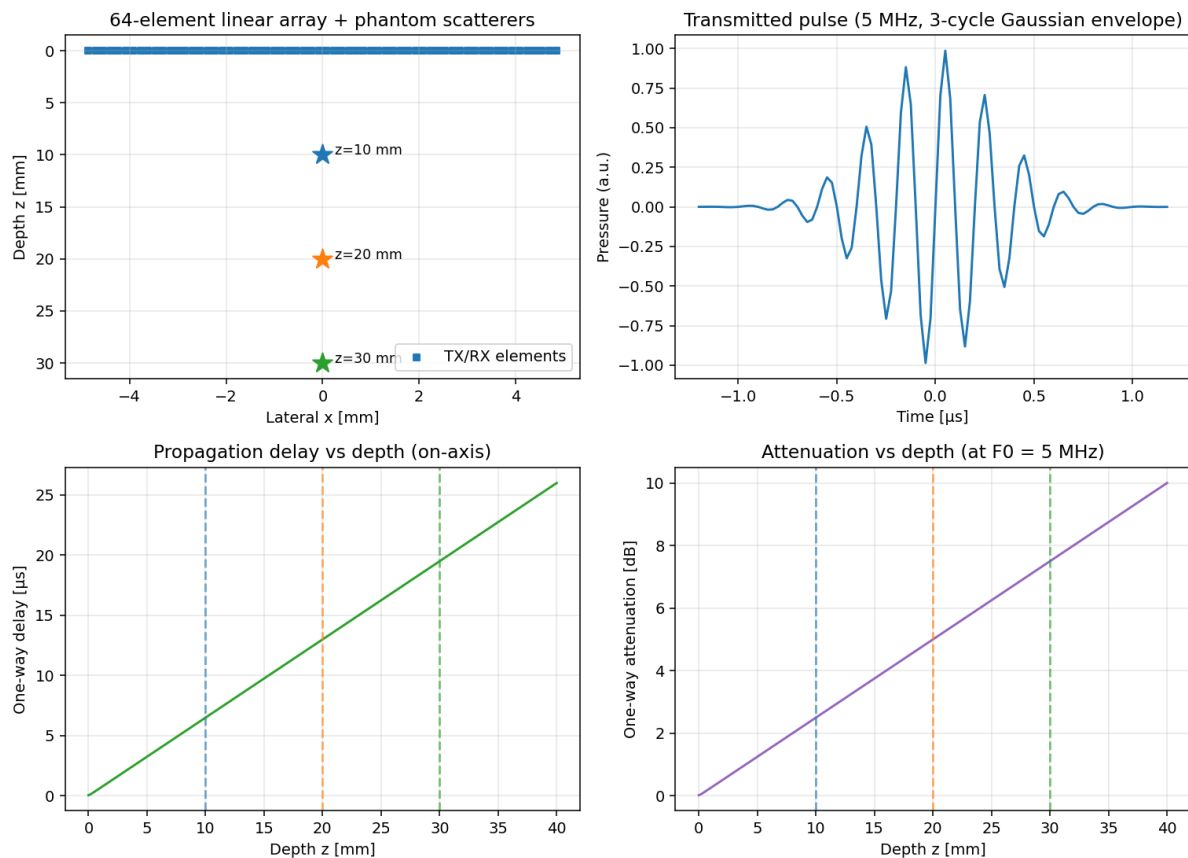
5 Output

Every figure and number produced by run_team3.py

5.1 Unit tests

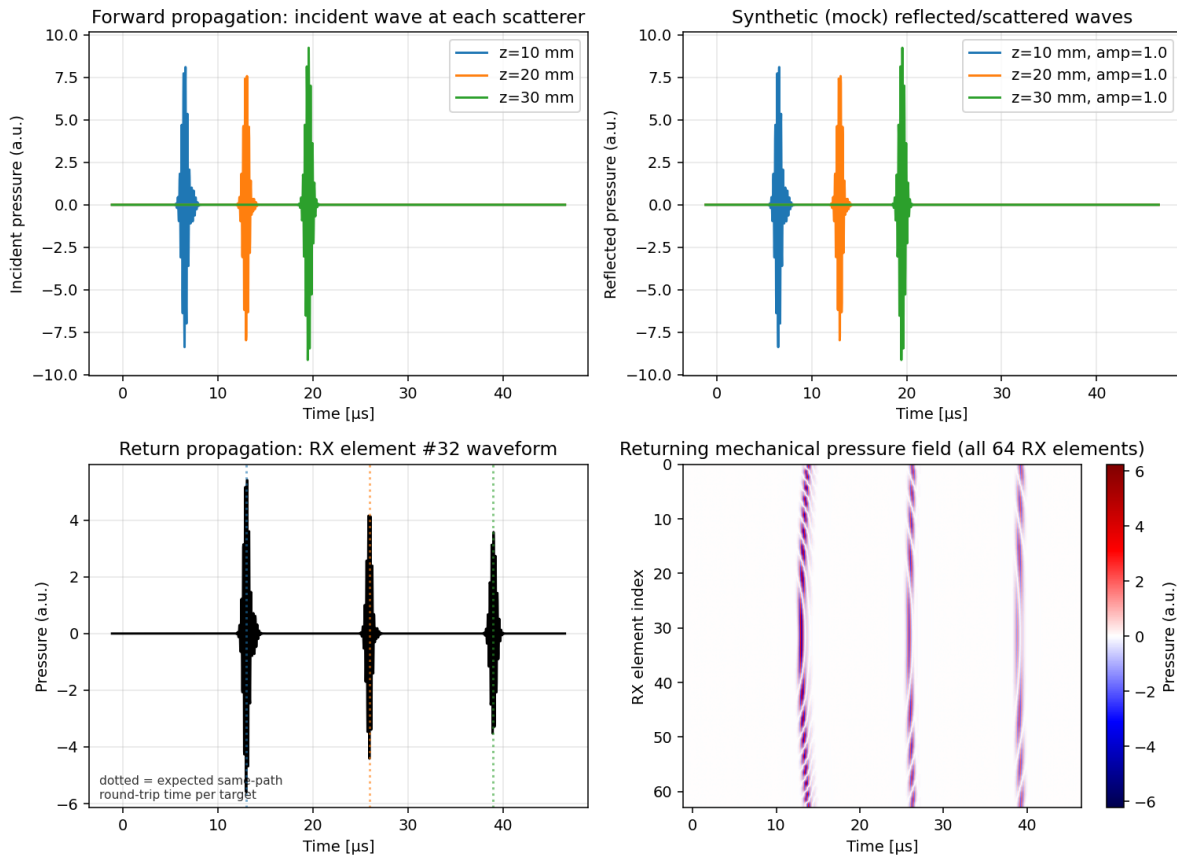
All **22 of 22** tests passed (Ran 22 tests ... 0K). The full unittest text is saved in output/unit_tests_full_output.txt.

5.2 Team demo, Figure 1: geometry, pulse, delay, attenuation



demo_geometry_pulse_delay_attenuation.png. Top left: the 64 elements (9.7 mm wide) and the three scatterers. Top right: the 5 MHz, 3-cycle pulse. Bottom left: one-way delay grows in a straight line with depth (about 6.5 us per 10 mm). Bottom right: one-way loss grows 2.5 dB per 10 mm at 5 MHz (5 dB at 20 mm).

5.3 Team demo, Figure 2: signals through the pipeline

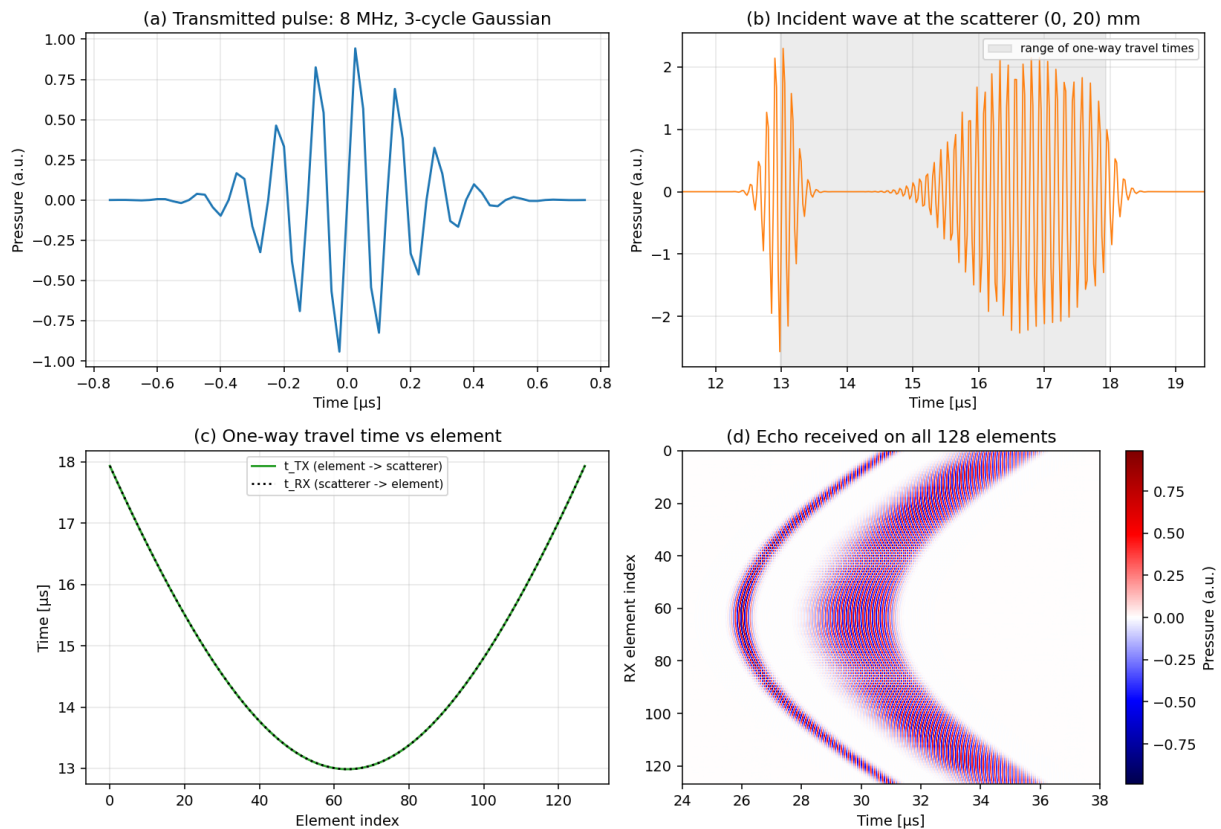


demo_pipeline_signals.png. Top left: incident wave at each scatterer, arriving at about 6.5, 13 and 19.5 μs . Top right: the reflected waves (identical, because $\text{amp} = 1.0$). Bottom left: the echo on element #32 lines up with the dotted expected round-trip times 12.99, 25.97 and 38.96 μs . Bottom right: all 64 elements; each echo is a slightly curved band because side elements are a little further away.

The two .npz files (part1_forward_propagation_outputs.npz and part2_return_propagation_outputs.npz) were compared with the copies the team saved earlier: same keys, and all numbers agree to within $2\text{e-}15$ (floating-point rounding only).

5.4 Final probe run (Philips L12-4 model)

Final probe run: Philips L12-4 model, point scatterer at (0, 20) mm



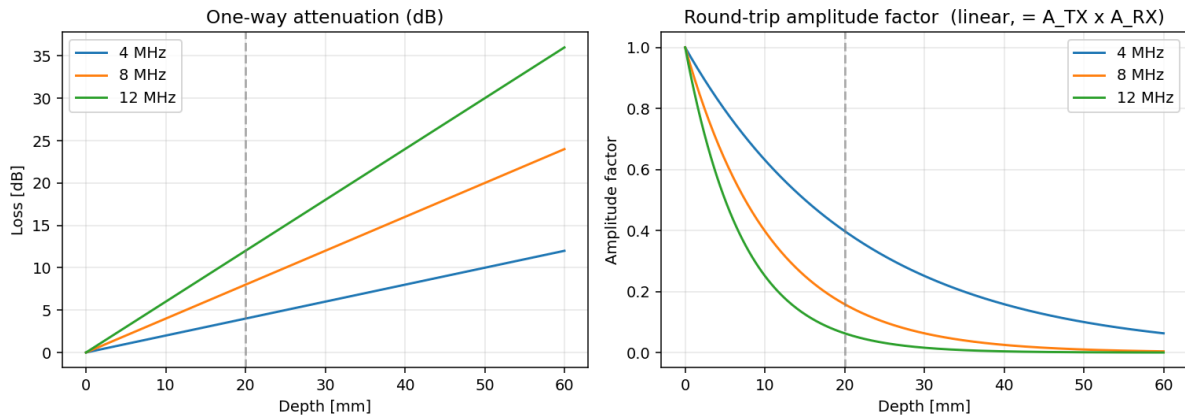
L12-4_final_probe_propagation.png. (a) the 8 MHz pulse (61 samples, ± 0.75 μs). (b) incident wave at (0, 20) mm: a strong burst at about 13 μs from the centre elements, then a longer train up to 17.94 μs from the outer elements (grey band = range of one-way times). (c) one-way time vs element: a U-shape from 12.99 μs (centre) to 17.94 μs (edges); t_{TX} and t_{RX} are identical. (d) echo on all 128 elements: every element receives the whole incident wave, delayed by its own t_{RX} , so the pattern is curved (earliest at the centre, 25.97 μs).

Quantity (from the console)	Value
Aperture / element x range	38.10 mm / -19.05 to 19.05 mm
One-way distance	20.001 to 27.621 mm
One-way time	12.99 to 17.94 μs
Round trip time	25.97 μs (centre) to 35.87 μs (edge)
One-way amplitude factor at 8 MHz	0.3981 (8.00 dB) to 0.2803 (11.05 dB)
Incident wave peak	2.565 a.u. at 12.97 μs
Received pressure array	(128, 1711) = elements x time samples
Peak on element #64 / #0	0.802 a.u. at 26.02 μs / 0.563 a.u. at 30.90 μs

Why is the incident wave so long?

No transmit focusing is used ($\tau_{tx} = 0$), so pulses from the 128 elements reach the scatterer at different times (12.99 to 17.94 μs). Near the centre the times are almost equal, so those pulses add up into one strong burst. Towards the edges each element arrives a bit later. A transmit beamformer (Team 2) would add delays τ_{tx} so all pulses arrive together; `forward_propagation()` already has the τ_{tx} argument for this.

5.5 Attenuation at 4, 8 and 12 MHz



attenuation_vs_depth_4_8_12MHz.png. Left: one-way loss in dB grows linearly with depth and with frequency. Right: round-trip amplitude factor ($A_{TX} \times A_{RX}$). At 20 mm (dashed line): 0.398 at 4 MHz, 0.159 at 8 MHz, 0.063 at 12 MHz. Higher frequency gives finer detail but less depth.

5.6 Console output

output/console_log.txt (exact text printed by python3 run_team3.py)

```
NITK-UtrasoundSim - Team 3: Acoustic Propagation - demo run
Python 3.12.4, numpy 1.26.4, matplotlib 3.8.3
Input file : input/team3_run_parameters.json
Output dir : output/

=====
STEP 1: Team 3 unit tests (test_acoustic_propagation.py)
=====
Command used: python -m unittest nitk_usoundsim.test_acoustic_propagation -v
(run inside the code/ folder)

ok    TestArrayGeometry.test_array_centred_on_zero
ok    TestArrayGeometry.test_element_count
ok    TestArrayGeometry.test_element_spacing_equals_pitch
ok    TestArrayGeometry.test_wavelength_and_pitch_relationship
ok    TestAttenuation.test_attenuation_at_20mm_is_5dB
ok    TestAttenuation.test_attenuation_is_bounded_between_0_and_1
ok    TestAttenuation.test_attenuation_matches_manual_db_formula
ok    TestAttenuation.test_deeper_targets_experience_more_attenuation
ok    TestDistancesAndDelays.test_deeper_targets_have_larger_tx_delay
ok    TestDistancesAndDelays.test_offaxis_target_gives_varying_rx_delays_across_elements
ok    TestDistancesAndDelays.test_rx_distance_matches_manual_formula
ok    TestDistancesAndDelays.test_rx_propagation_delay
ok    TestDistancesAndDelays.test_tx_distance_matches_manual_formula
ok    TestDistancesAndDelays.test_tx_distance_on_axis_center_equals_depth
ok    TestDistancesAndDelays.test_tx_propagation_delay
ok    TestFullPipeline.test_deeper_phantom_points_have_larger_min_delay
ok    TestFullPipeline.test_multiple_scatterer_linear_superposition
ok    TestFullPipeline.test_output_dimensions
ok    TestFullPipeline.test_output_is_real_valued_mechanical_pressure
ok    TestFullPipeline.test_round_trip_delay_for_20mm_phantom_point
ok    TestPulse.test_pulse_centered_near_zero
ok    TestPulse.test_pulse_sampled_at_fs
Ran 22 tests in 0.023s
OK
(full unittest text saved in output/unit_tests_full_output.txt)

RESULT: 22 of 22 tests passed -> ALL PASSED

=====
STEP 2: Team demo - 64 elements, 5 MHz, lambda/2 pitch (config.py)
=====
Input check: values in input JSON match config.py -> True
Elements      : 64
Centre freq   : 5.0 MHz
```

```

Sampling freq : 40.0 MHz
Speed of sound: 1540 m/s
Wavelength    : 0.3080 mm
Pitch (lam/2) : 0.1540 mm
Attenuation    : 0.5 dB/MHz/cm
Scatterers     : (0, 10) mm, (0, 20) mm, (0, 30) mm

Saved figure: output/demo_geometry_pulse_delay_attenuation.png
Saved figure: output/demo_pipeline_signals.png
Saved PART 1 (forward propagation) data: output/part1_forward_propagation_outputs.npz
Saved PART 2 (return propagation) data: output/part2_return_propagation_outputs.npz

Time axis      : 1910 samples, -1.175 to 46.550 us
PART 1 incident_pressure shape : (3, 1910) (scatterers, time)
PART 2 received_pressure shape : (64, 1910) (RX elements, time)
Per scatterer (on-axis, round trip = min t_TX + min t_RX):
  z = 10 mm: one-way t = 6.494- 7.217 us | round trip = 12.987 us (2z/c = 12.987) | 2-way amp = 0.562
  z = 20 mm: one-way t = 12.987-13.364 us | round trip = 25.974 us (2z/c = 25.974) | 2-way amp = 0.316
  z = 30 mm: one-way t = 19.481-19.734 us | round trip = 38.961 us (2z/c = 38.961) | 2-way amp = 0.178

=====
STEP 3: Final probe - Philips L12-4 model (128 el, 0.30 mm, 8 MHz)
=====
Done ONLY by passing arguments to the team's functions (code unchanged).
  Elements      : 128 pitch = 0.30 mm aperture = 38.10 mm
  Element x range : -19.05 to 19.05 mm
  Pulse         : 8 MHz, 3 cycles, sigma = 0.1875 us, half-width = 0.7500 us, 61 samples
  Scatterer      : (0, 20) mm, amp = 1.0
  Time axis      : 1711 samples at 40 MHz, -0.750 to 42.000 us
  One-way distance: 20.001 to 27.621 mm
  One-way time    : 12.99 to 17.94 us (centre to edge elements)
  Round trip time : 25.97 us (centre element) to 35.87 us (edge element)
  One-way amplitude factor at 8 MHz: 0.3981 (centre) to 0.2803 (edge) = 8.00 to 11.05 dB
  Incident wave peak |p| = 2.565 a.u. at t = 12.97 us (sum of 128 attenuated pulses)
  Received pressure array shape: (128, 1711) (RX elements, time)
  Element #64: peak |p| = 0.802 a.u. at t = 26.02 us
  Element #0 : peak |p| = 0.563 a.u. at t = 30.90 us

Saved figure: output/L12-4_final_probe_propagation.png
Saved data : output/L12-4_final_probe_outputs.npz

=====
STEP 4: Attenuation vs depth at 4, 8, 12 MHz (calculate_attenuation)
=====
alpha0 = 0.5 dB/MHz/cm. One-way and two-way (round trip) loss:
  f0      depth 20 mm: one-way dB  amp factor | round-trip dB  amp factor
    4 MHz           4.00      0.6310 |      8.00      0.3981
    8 MHz           8.00      0.3981 |     16.00      0.1585
   12 MHz          12.00      0.2512 |     24.00      0.0631
Saved figure: output/attenuation_vs_depth_4_8_12MHz.png

=====
SUMMARY
=====
Unit tests : 22/22 passed
Output files in output/:
- L12-4_final_probe_outputs.npz
- L12-4_final_probe_propagation.png
- attenuation_vs_depth_4_8_12MHz.png
- demo_geometry_pulse_delay_attenuation.png
- demo_pipeline_signals.png
- part1_forward_propagation_outputs.npz
- part2_return_propagation_outputs.npz
- unit_tests_full_output.txt
- console_log.txt
- console_screenshot.png
Total run time: 1.5 s

```

```

● ● ● python3 run_team3.py (Team 3: Acoustic Propagation)

NITK-UsoundSim - Team 3: Acoustic Propagation - demo run
Python 3.12.4, numpy 1.26.4, matplotlib 3.8.3
Input file : input/team3_run_parameters.json
Output dir : output/

=====
STEP 1: Team 3 unit tests (test_acoustic_propagation.py)
=====
Command used: python -m unittest nitk_usoundsim.test_acoustic_propagation -v
(run inside the code/ folder)

ok TestArrayGeometry.test_array_centred_on_zero
ok TestArrayGeometry.test_element_count
ok TestArrayGeometry.test_element_spacing_equals_pitch
ok TestArrayGeometry.test_wavelength_and_pitch_relationship
ok TestAttenuation.test_attenuation_at_20mm_is_5dB
ok TestAttenuation.test_attenuation_is_bounded_between_0_and_1
ok TestAttenuation.test_attenuation_matches_manual_db_formula
ok TestAttenuation.test_deeper_targets_experience_more_attenuation
ok TestDistancesAndDelays.test_deeper_targets_have_larger_tx_delay
ok TestDistancesAndDelays.test_offaxis_target_gives_varying_rx_delays_across_elements
ok TestDistancesAndDelays.test_rx_distance_matches_manual_formula
ok TestDistancesAndDelays.test_rx_propagation_delay
ok TestDistancesAndDelays.test_tx_distance_matches_manual_formula
ok TestDistancesAndDelays.test_tx_distance_on_axis_center_equals_depth
ok TestDistancesAndDelays.test_tx_propagation_delay
ok TestFullPipeline.test_deeper_phantom_points_have_larger_min_delay
ok TestFullPipeline.test_multiple_scatterer_linear_superposition
ok TestFullPipeline.test_output_dimensions
ok TestFullPipeline.test_output_is_real_valued_mechanical_pressure
ok TestFullPipeline.test_round_trip_delay_for_20mm_phantom_point
ok TestPulse.test_pulse_centered_near_zero
ok TestPulse.test_pulse_sampled_at_fs
Ran 22 tests in 0.023s
OK
(full unittest text saved in output/unit_tests_full_output.txt)

RESULT: 22 of 22 tests passed -> ALL PASSED

=====
STEP 2: Team demo - 64 elements, 5 MHz, lambda/2 pitch (config.py)
=====
Input check: values in input JSON match config.py -> True
Elements : 64
Centre freq : 5.0 MHz
Sampling freq : 40.0 MHz
Speed of sound: 1540 m/s
Wavelength : 0.3080 mm
Pitch (lam/2) : 0.1540 mm
Attenuation : 0.5 dB/MHz/cm
Scatterers : (0, 10) mm, (0, 20) mm, (0, 30) mm

Saved figure: output/demo_geometry_pulse_delay_attenuation.png
Saved figure: output/demo_pipeline_signals.png
Saved PART 1 (forward propagation) data: output/part1_forward_propagation_outputs.npz
Saved PART 2 (return propagation) data: output/part2_return_propagation_outputs.npz

Time axis : 1910 samples, -1.175 to 46.550 us
PART 1 incident pressure shape : (3, 1910) (scatterers, time)
PART 2 received pressure shape : (64, 1910) (RX elements, time)
Per scatterer (on-axis, round trip = min t_TX + min t_RX):
z = 10 mm: one-way t = 6.494- 7.217 us | round trip = 12.987 us (2z/c = 12.987) | 2-way amp = 0.562
z = 20 mm: one-way t = 12.987-13.364 us | round trip = 25.974 us (2z/c = 25.974) | 2-way amp = 0.316
z = 30 mm: one-way t = 19.481-19.734 us | round trip = 38.961 us (2z/c = 38.961) | 2-way amp = 0.178

=====
STEP 3: Final probe - Philips L12-4 model (128 el, 0.30 mm, 8 MHz)
=====
Done ONLY by passing arguments to the team's functions (code unchanged).
Elements : 128 pitch = 0.30 mm aperture = 38.10 mm
Element x range : -19.05 to 19.05 mm
Pulse : 8 MHz, 3 cycles, sigma = 0.1875 us, half-width = 0.7500 us, 61 samples
Scatterer : (0, 20) mm, amp = 1.0
Time axis : 1711 samples at 40 MHz, -0.750 to 42.000 us
One-way distance: 20.001 to 27.621 mm
One-way time : 12.99 to 17.94 us (centre to edge elements)
Round trip time : 25.97 us (centre element) to 35.87 us (edge element)
One-way amplitude factor at 8 MHz: 0.3981 (centre) to 0.2803 (edge) = 8.00 to 11.05 dB
Incident wave peak |p| = 2.565 a.u. at t = 12.97 us (sum of 128 attenuated pulses)
Received pressure array shape: (128, 1711) (RX elements, time)
Element #64: peak |p| = 0.802 a.u. at t = 26.02 us
Element #0 : peak |p| = 0.563 a.u. at t = 30.90 us

Saved figure: output/L12-4_final_probe_propagation.png
Saved data : output/L12-4_final_probe_outputs.npz

=====
STEP 4: Attenuation vs depth at 4, 8, 12 MHz (calculate attenuation)
=====
alpha0 = 0.5 dB/MHz/cm. One-way and two-way (round trip) loss:
f0 depth 20 mm: one-way dB amp factor | round-trip dB amp factor
4 MHz 4.00 0.6310 | 8.00 0.3981
8 MHz 8.00 0.3981 | 16.00 0.1585
12 MHz 12.00 0.2512 | 24.00 0.0631
Saved figure: output/attenuation_vs_depth_4_8_12MHz.png

=====
SUMMARY
=====
Unit tests : 22/22 passed
Output files in output/:
- L12-4_final_probe_outputs.npz
- L12-4_final_probe_propagation.png
- attenuation_vs_depth_4_8_12MHz.png
- demo_geometry_pulse_delay_attenuation.png
- demo_pipeline_signals.png
- part1_forward_propagation_outputs.npz
- part2_return_propagation_outputs.npz
- unit_tests_full_output.txt
- console_log.txt
- console_screenshot.png
Total run time: 1.5 s

```

output/console_screenshot.png: the same console text as an image.

6 How to run

Step-by-step, on any computer

1. Install Python 3 (3.9 or newer) from python.org. On Windows, tick *Add Python to PATH* during installation.
2. Copy the folder `Team_3_Acoustic_Propagation` anywhere you like.
3. Open a terminal in that folder (macOS: right-click the folder in Finder, *New Terminal at Folder*; Windows: open the folder, type `cmd` in the address bar, press Enter).
4. Install the packages: `pip install -r requirements.txt` (macOS/Linux may need `pip3` or `python3 -m pip`).
5. Run: `python3 run_team3.py` (macOS/Linux) or `python run_team3.py` (Windows).
6. Open the `output/` folder to see the plots, data, `console_log.txt` and `console_screenshot.png`.

Expected runtime: measured 2 to 5 seconds in total (the script itself reports about 1.5 to 2 s) on the test Mac.

Running the team's files one by one

`run_team3.py` runs everything at once. `nitk_usoundsim` is a Python package, so its files are run from the `code/` folder (the team's docstrings call it the *usound_sim* folder):

Optional, from the code/ folder

```
cd code
python3 -m nitk_usoundsim.demo_acoustic_propagation
python3 -m unittest nitk_usoundsim.test_acoustic_propagation -v
python3 nitk_usoundsim/check_arr.py
```

- The demo saves its two figures and the `part1/part2 .npz` files in `code/nitk_usoundsim/`, overwriting the earlier copies. `run_team3.py` avoids that by writing to `output/`.
- The unit test command runs the 22 tests.
- `check_arr.py` prints the arrays in `part2_return_propagation_outputs.npz`. It opens that file by the path `nitk_usoundsim/...`, so it must be started from `code/` (the VS Code Run button starts it from the wrong folder).
- In VS Code: open this team folder, pick a Python interpreter with the packages from `requirements.txt`, open a file and press Run. `.vscode/settings.json` adds `code/` to the import path in the VS Code terminal.

Troubleshooting

Problem	Fix
<code>python3 / python: command not found</code>	Install Python, or on Windows try <code>py run_team3.py</code> .
<code>ModuleNotFoundError: numpy or matplotlib</code>	Run <code>python3 -m pip install -r requirements.txt</code> with the same Python.
<code>ModuleNotFoundError: nitk_usoundsim</code>	Run the command from inside the <code>Team_3_Acoustic_Propagation</code> folder and keep the <code>code/</code> folder next to <code>run_team3.py</code> .
No plots appear on screen	That is normal: plots are saved as PNG files in <code>output/</code> , not shown in a window.
Permission error writing <code>output/</code>	Copy the folder to a place where you can write (e.g. Desktop or Documents).

7 Check yourself

Viva-style questions with short answers

Q1. How long does sound take to reach a scatterer 20 mm straight below the probe, and to come back?

Answer: $t = d / c = 0.020 / 1540 = 12.99 \text{ us}$ one way, 25.97 us round trip.

Q2. Why does the attenuation formula use $10^{(-dB/20)}$ and not $10^{(-dB/10)}$?

Answer: Decibels are defined on power, and power goes with pressure squared. The code works with pressure amplitude, so it uses 20.

Q3. Why do edge elements of the 128-element probe receive the echo later than the centre element?

Answer: They are further from the scatterer (27.62 mm vs 20.00 mm by Pythagoras), so the one-way time grows from 12.99 us to 17.94 us.

Q4. Why does `forward_propagation` use the pulse formula, but `return_propagation` uses `np.interp`?

Answer: The transmitted pulse is a formula, so it can be evaluated exactly at any delayed time. The reflected wave is only known as samples, so values between samples are found by linear interpolation.

Q5. Is 40 MHz sampling enough for an 8 MHz pulse?

Answer: Yes. Nyquist needs more than twice the highest frequency; the pulse energy is well below 20 MHz, and there are 5 samples per 8 MHz period.

Q6. Name two simplifications of this model and what would be needed to remove them.

Answer: E.g. point elements (add element width/directivity), no 1/r spreading (divide by distance), constant c (allow different tissues), mock tissue (replace with Team 4's real tissue interaction).